

How archival eclipse timing can produce period change

Three mechanisms quantified on 26 short-period binaries

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ABSTRACT

Context. A quadratic ephemeris fitted to decades of eclipse timings returns a period-change coefficient whatever the data contain.

Aims. To quantify the three mechanisms on one sample, and to test the resulting coefficients against the complete TESS record of the same targets.

Methods. Timings of 26 binaries from TESS 2-min photometry and SuperWASP (2004–2008), with bars from the data; a residual test with its null simulated as the pipeline operates; injected orbits and secular terms; a between-sector noise model measured on 799 sectors.

Results. With formal errors the chain makes 24 of 31 targets “significant”. With bars co-estimated from two or three samples the residual test has a null of 0.77, not 1, and cannot separate correct bars from bars inflated by a real secular term, whose $\sigma(dP/dt)$ it then overestimates by a median factor of 2.9 at 0.2 s yr^{-1} . With data-derived bars 11 of 26 designs pass an adversarial gate, and 7 survive a noise model admitted by its own goodness of fit at the between-sector scale. The complete TESS block — 38 to 43 sectors per target, 35 to 39 of them independent of the design — **contradicts four of the seven coefficients it classifies** at more than 3σ under every admitted noise model and leaves one ambiguous; of the two it does not contradict, TIC 230386284 agrees at a power of 0.0024 s yr^{-1} and the other is untested. V527 Dra, with a published light-travel-time orbit, is held out and its complete fit is compatible with zero. Undersampled orbits account for $N = 1.25$ (95% 0.80–2.46) of them, and the archival timing bias, recalibrated against ground-based ephemerides, is $-1.26 \pm 0.75 \text{ min}$.

Conclusions. A quadratic coefficient fitted over 16–20 yr to a few epochs is a coefficient of its window: the complete block contradicts four of the seven it classifies and agrees, with power, with one.

Key words. binaries: eclipsing – stars: fundamental parameters – methods: data analysis – methods: statistical – techniques: photometric – ephemerides

1. Introduction

Secular period changes in close binaries encode mass transfer, angular-momentum loss and light-travel-time effects, but measuring them requires eclipse timings spanning decades, and the archival photometry that supplies the baseline comes with heterogeneous time systems, unquantified timing biases and no ground truth internal to the archive. A quadratic ephemeris fitted to a few epochs over such a baseline returns a coefficient whatever the data contain. This paper is about the ways that coefficient can be produced, and about how to tell.

Three mechanisms are quantified on one uniformly processed sample: timing errors that are formal rather than measured; timing bars that are estimated from too few samples and applied at another time-scale, where the question of *whether* a bar is wrong is answered by a leverage-corrected decomposition of the residuals against the null of its own estimator; and third-body orbits sampled at a few phases, which a parabola absorbs into a coefficient that every internal test then certifies. The chain was developed on one system (TYC 7024-1046-1 = TIC 142874476), where a SuperWASP epoch from 2006–2008 propagated through a TESS-refined ephemeris sits $21.8 \pm 2.2 \text{ min}$ from the linear prediction, and is applied here to every catalogued binary for which the data allow the tests to run. The 26 binaries are the vehicle, not the object: they are the subset

with data sufficient to demonstrate the mechanisms, drawn from a cache rather than from the sky (Sect. 2), and no population statement is made from them.

None of the three mechanisms is unknown. That formal photon-noise errors underestimate eclipse-timing uncertainties is why timing studies use the scatter of the data; that a light-travel-time orbit sampled over a fraction of its period is degenerate with a quadratic ephemeris is a standing caveat of the eclipse-timing literature, stated wherever a cyclic and a secular term are fitted to the same O–C (e.g. Borkovits et al. 2016; Hajdu et al. 2019), and the timing of eclipsing binaries from TESS itself has been examined for exactly this degeneracy (Marcadon & Prša 2024). What is new here is not the mechanisms but their measurement on one sample with one chain: a detection efficiency for undersampled orbits measured target by target at the real epochs with the real bars; one target with a published orbit and no secular term (V527 Dra) in which every internal test — significance, adversarial, goodness of fit — passes the confounder; a leverage-corrected decomposition of the residuals, with the null distribution of the estimator obtained by re-estimating the bars inside the simulation, which says whether either archive carries excess variance and — in this sample — traces an apparent archival excess to one estimation error measured against the wrong null and the remainder to three targets, rather than to the archive; and a test of the design inside the sample, in which the

Table 1. Selection funnel.

Step	Criterion	N
1	Union of the TESS eclipsing-binary catalogues of Prša et al. (2022; 4 584) and Kostov et al. (2025; the 7 936 systems new to that catalogue, its table 3; the 2 065 previously known systems of its table 4 are not used), de-duplicated by position (5'') over the merged table	12 497
2	Catalogue period $P < 5$ d	8 759
3	≥ 2 TESS 2-min sectors separated by ≥ 1 yr among the 22 sector manifests held locally (s19–21, 27–28, 75, 90–105)	288
4	A SuperWASP source within 20'' (public archive cone search — the query that lists sources, not the light-curve download; 0 of 288 queries unresolved after retry)	131
5	≥ 4 independent epoch clusters (TESS sectors + SuperWASP seasons), the minimum for a quadratic ephemeris with ≥ 1 degree of freedom	88
6	Passed all measurement guards (Sect. 3)	26

coefficient fitted to four epochs is confronted with the one the complete TESS block of the same target gives.

2. The sample as a vehicle: 26 binaries with data sufficient for the tests

Selection is deterministic and every cut is listed with its survivors (Table 1). The sample exists to run the tests of Sect. 5, not to represent a population; the targets that were *not* measured are nevertheless reported with the reason (Table 2), because the guards that stopped them are part of the method.

Two remarks. The two catalogues are disjoint by construction (0 TICs in common; 23 positional pairs at 5''), so the union genuinely adds. And **the sector count of step 3 is a measure of our cache, not of the sky**: the 22 local manifests cover 21% of TESS sectors, so 288 is a lower bound, and a biased one.

The cache census. The bias was measured rather than left as a caveat (src/vies_cache_288.py; expectation written before running: a median ratio of 2–3 and 15–25% of targets at a ratio ≥ 5). For each of the 288 targets that passed step 3, MAST was asked how many SPOC 2-min sectors exist (288 of 288 answered, 0 network failures). The cache holds a median of 4 sectors per target against an archive median of 32 (2–45), 7 857 sectors against 1 224; the ratio has median 5.7, is ≥ 5 for 62% of the 288, and rises with ecliptic latitude (Spearman +0.60, $p = 5 \times 10^{-30}$), from 2.3 at $|\beta| < 40^\circ$ to 10.0 at $78\text{--}90^\circ$, where half of the 288 sit. The 26 measured targets are the extreme of this: ratio 10.5. The expectation was wrong by a factor of two in the median and three in the fraction above 5; the undercount is severe everywhere and worst exactly where the sample lives. Those 799 light curves are downloaded and used as the test of Sect. 5.5.

What this implies is stated once and used throughout: this is not a sample of short-period binaries, it is a sample of binaries in the continuous-viewing zones with deep archival photometry — which is what makes it adequate as a vehicle for the tests of Sect. 5 and inadequate as a population. The sky distribution of the result follows directly: 21 of the 26 measured targets are in Draco (the others: 2 in Microscopium, 2 in Grus, 1 in Indus), be-

cause the cached northern sectors overlap there and SuperWASP covered it in 2004–2008. The sample is a Draco sample with a southern tail, not an all-sky one — which is why no population claim is made from it (Sect. 6).

The design and the complete block. The chain fits the design of four to seven epochs per target: the sectors already in the local cache — a median of four per target — plus the SuperWASP seasons. The 799 2-min light curves that MAST holds for the 26 are also used here, but in a different role: they are not folded into the design, they *test* it. Keeping the design fixed is what makes the tests of Sect. 5 comparable across versions of the chain, and four sectors per target is the regime in which the archival seasons carry the coefficient — the regime this paper is about; the complete block, 35 to 43 primary epochs per target over 2019.6–2024.9, gives an independent estimate of the same coefficient over the TESS window alone (Sects. 5.3.2 and 5.5). The epochs of the design are a subset of the complete block and are measured identically, so removing them from the block leaves a set that shares no epoch with the design: the test is repeated that way, and the classes it returns do not change (Sect. 5.5).

Where the block ends, and why. The complete block stops at sector 86 (2024-12-18) for all eight targets that carry the test, and this is a pointing effect, not a cache effect: a MAST inventory of every time-series product of the eight (136–188 products each) returns nothing after s86, because the eight are in Draco (Dec $+56^\circ$ to $+64^\circ$) and TESS observed the south in 2025–2026 — the control target at Dec -34° has sectors 97 and 105. An out-of-sample test with sectors later than the design is therefore not available today; its per-class prediction is registered in advance (Sect. 5.5). For when those sectors arrive, the two full-frame-image reductions were compared with the 2-min epochs of the same sectors (s84–s86, 38 measurements, 200-s cadence): TESS-SPOC agrees with a median $|t(\text{FFI}) - t(2\text{-min})|$ of 0.30 min and no systematic offset (-0.06 ± 0.16 min), while QLP does not (differences up to 7.2 min, and in s85 a shift of the same sign in all seven targets with a formal bar that does not cover it). TESS-SPOC 200-s photometry is usable for this timing; QLP needs a per-sector check against 2-min data.

A recovery pass was run after the main batch on the two rows a local action could change (intermediate TESS sectors for the ladder row; eclipse durations measured on the TESS curves for the duration row) and recovered none of them: behind the first guard there was a second in every case, so recovery is not additive across the rows of Table 2 and the count of 26 stands (Appendix A). The 26 are therefore not a random sample of short-period binaries: they are the subset with deep SuperWASP coverage, eclipses sampled in every season, and unambiguous cycle counts across the sectors held locally.

3. Method

3.1. Eclipse epochs

For each TESS sector we fit a periodic trapezoid of free depth to the PDCSAP light curve (QUALITY = 0, normalised to the 95th percentile) on a fine grid of trial epochs, with the eclipse duration T_{14} from the catalogue and the period from the catalogue, seeded on the catalogue reference epoch so that the primary is fitted rather than the deepest minimum in the curve (in contact systems these differ). The epoch is the χ^2 minimum; the formal error is the $\Delta\chi^2 = 1$ half-width. Sectors are anchored to the most recent one.

Table 2. The 62 targets of the 88 that were not measured, by the first guard each one failed. This table is a record of where each target stopped, not a map of what recovery would gain: every target fails at the first guard it meets, and behind the first there is usually another (see the recovery pass below). The “Nature” column describes the first reason only.

Reason	N	Nature
SuperWASP curve with < 500 valid points after cleaning	22	archive depth; structural
Phase coverage of the eclipse not testable in ≥ 1 season block (Sect. 3.3; a season whose fitted depth is ≤ 0 counts as not testable, Sect. 3.1)	22	archive sampling; recoverable with a second archive
SuperWASP light-curve download returned an empty file in four attempts (a different operation from the cone search of Table 1, step 4)	4	archive-side failure; recoverable if the archive serves the file
Cycle count between TESS sectors ambiguous (period ladder did not close)	8	local sector coverage; intermediate sectors were added from MAST and none was recovered (Appendix A)
No applicable eclipse duration in the catalogue (4 NaN; 1 contact-binary width; 1 W UMa)	6	catalogue gap; T_{14} was measured on the TESS curve for five and none was recovered (Appendix A)

For SuperWASP we convert HJD_UTC to BJD_TDB (a +1.1 min shift at these dates — larger than the TESS timing error), fit the same profile per observing season, and require phase coverage of the eclipse in every season (≥ 50 points in eclipse, no empty phase bins in 12). A season whose fitted depth is ≤ 0 is rejected: the grid has found a χ^2 minimum for a *brightening*, and its “epoch” is not that of an eclipse. Five of the first 31 targets had such a season; their between-season dispersion was 82 min against 1.8 min for the rest.

Blending. A constant neighbour in the aperture dilutes the depth of an eclipse and does not move its time of minimum; only a neighbour that is itself variable on the time-scale of the eclipse can shift a minimum or add timing structure. What can be measured cheaply is who has a neighbour capable of it (src/blending_26.py; expectation written and committed before running). The SPOC crowding metric CROWDSAP of the 84 sectors used has a median of 0.97 per target (median over its sectors), 14 of the 84 sectors are below 0.9 and none below 0.5; the lowest is TIC 377253090 (0.56 in all four sectors). Gaia DR3 sources within 48'' (about two TESS pixels) and within 2 mag in G of the target exist for three of the 26: TIC 229476285 (18.8'', $\Delta G = 1.5$), TIC 359552377 (10.1'', $\Delta G = 1.9$), and TIC 377253090, a member of D9, which has a companion of nearly equal brightness at 1.7'' ($\Delta G = 0.37$) that neither TESS nor SuperWASP resolves: which of the two stars is the eclipsing binary, and whether the other is variable, is not decided by the data of this paper, and the detection is carried with that caveat. No blending correction is applied to any epoch, since none is needed for a constant neighbour; the variability of the neighbours was not tested.

3.2. Period ladder and cycle count

Within the TESS sectors the period is refined by a ladder: starting from the two closest sectors, each further sector is added only if its linear residual stays below a tolerance equal to the larger of 3 min and three times the median epoch uncertainty of the target (3 min for 22 of the 26; 3.2–5.5 min for the other 4), and the result is discarded if shifting any epoch by ± 1 cycle does not worsen χ^2 . The ladder returns epochs and residuals in the caller’s order (Sect. 5).

The ladder is *not* used across the 2008–2019 gap: it requires a linear ephemeris, and a real dP/dt produces tens of minutes of O–C over 19 yr, which the ladder would reject as an unresolved alias. Across the gap the cycle number is the rounded propagation with the TESS-refined period, and the count is accepted

only if $\sigma_P \times N < 0.25$ cycle at the farthest epoch. Any target with $|dP/dt| > 10 \text{ s yr}^{-1}$ is flagged as a suspected cycle-count error rather than a measurement; none of the 26 is.

3.3. Timing uncertainties from the data

The formal $\Delta\chi^2 = 1$ error is a photon-noise error. Cycle-to-cycle eclipse-timing scatter from spots and O’Connell-type asymmetries is not in it, and in the SuperWASP data the dispersion between observing seasons exceeds the formal error by a median factor of 4.3 ($n = 25$ targets of the 26; one record lacks the formal value; interquartile range of the factor 2.9–10.7). We therefore assign each epoch the *larger* of its formal error and the dispersion between independent subsets: the two halves of the sector for TESS, the observing seasons for SuperWASP. Each SuperWASP season enters the fit as its own epoch with this error — the dispersion of one season, $\text{std}(d)$, not the standard error of the season mean, $\text{std}(d)/\sqrt{n}$, which is the bar of the *combined* epoch used on the control target; an earlier version assigned the latter to each season (Sect. 5.2.2, Appendix D) — added in quadrature to the chain-bias uncertainty of Sect. 3.4. A target with a single valid season has no red-noise estimate and is not measured. Where the archive holds one observing season only, the season is split at its midpoint into two blocks of 39–43 nights and the bar is the dispersion between the halves (as for TESS): this is the case for five of the 26 (TIC 229461186, 229476285, 229914020, 230386284 and 424461577, all with two 2008 blocks 0.11–0.12 yr apart), whose “two seasons” in Table 3 are two halves of one, and whose archival lever is a single epoch at 2008.5 rather than two seasons years apart. Two properties of these bars are shown by the data: because each is estimated from two or three samples that also enter the residual, the residual test has a null below 1 that must be simulated rather than assumed (Sect. 5.2.3), and the TESS bar, being a within-sector quantity, cannot contain timing drift between sectors (Sect. 5.3.2). For TESS the bar is $\max(\text{formal}, |t_a - t_b|/2)$, with t_a and t_b the epochs fitted to the two halves of the sector: each half has an error $\sqrt{2}$ larger than the whole sector, so the difference of the two, divided by 2, estimates the error of the whole-sector epoch (a simulation of two halves with known scatter recovers the whole-sector error with $|t_a - t_b|/2$ and overestimates it by $\sqrt{2}$ with $|t_a - t_b|/\sqrt{2}$). The halves term sets the bar in 22 of the 84 TESS epochs and the formal error in the other 62. The divisor was $\sqrt{2}$ in an earlier version — the dispersion of *one half* given to the whole sector; the correction and what it moved are in Appendix D. The control

target does not use the halves rule (its bar is the spread between three reductions; Appendix B) and reproduced before and after.

3.4. SuperWASP timing bias

There is no ground truth internal to the archive, so the timing bias of the SuperWASP chain is calibrated on transiting hot Jupiters — objects whose clocks do not run. Eclipsing binaries cannot serve: a first attempt with three gave an astrophysical O–C scatter of 31.7 min against 1.73 min for the hot Jupiters. Of 12 hot Jupiters, 8 yielded epochs and 4 passed the between-season dispersion guard (WASP-18 b, WASP-4 b, WASP-19 b and WASP-6 b, in the order used below).

The calibration used here compares those four seasons with **ephemerides that include ground-based timings** rather than with a TESS-only propagation (`src/vies_hj_indep.py`): the homogeneous transit-timing ephemerides of Ivshina & Winn (2022) and the ExoClock III ephemerides of Kokori et al. (2023), and for WASP-4 b the quadratic ephemeris of Bouma et al. (2020), whose published period derivative is propagated with its own uncertainty (the covariance is not published, so the propagation is diagonal). The four O–C are -2.38 ± 1.08 (WASP-18 b), -0.81 ± 1.45 (WASP-4 b), $+0.46 \pm 2.59$ (WASP-19 b) and $+0.40 \pm 1.85$ min (WASP-6 b), with the ephemerides of Ivshina & Winn (2022) and the Bouma et al. (2020) term on WASP-4 b, consistent with a constant offset ($\chi^2 = 2.4$ for 3 d.o.f.):

$$\text{chain bias} = -1.26 \pm 0.75 \text{ min.} \quad (1)$$

The two ephemeris sources agree to 0.03–0.40 min per season and give weighted means from -1.26 to -1.56 min; the propagated ephemeris uncertainties are 0.05–0.29 min, negligible against the 1.0–2.6 min of the archival seasons. The value is $1.7\text{--}2.1\sigma$ from zero. A TESS-only propagation gives -2.14 ± 0.78 min, and the difference is almost entirely WASP-4 b, whose published orbital decay the TESS-only propagation folds into the bias (-2.55 min there, -0.81 min here).

Three candidate mechanisms for a bias of this size were tested and none reproduces it. (i) *Shape*. A procedural bias arising in the fitting, the sampling or the profile is shape-dependent, and the four calibrators are shallow, long transits while the targets are deep, sharp eclipses. Transits of the published geometry were injected into the real SuperWASP curves of the four and measured by the same chain (`src/vies_forma_swasp.py`): the weighted O–C is -0.10 min (95% interval -2.8 to $+2.4$), i.e. the offset of the earlier, TESS-only calibration is not reproduced by the estimator acting on the transit shape. The same injection on the eclipse shapes of the 26 gives per-target biases of up to ± 3 min, driven by template mismatch combined with asymmetric phase coverage — so a *common* bias is not what the estimator produces, while a *per-target* one of a few minutes is. (ii) *Time system*. Twenty raw archival timestamps were converted independently with *astropy* and compared with what the chain produces (`src/checa_tempo_swasp.py`): the two agree to 0.002 s on average and 0.019 s at worst, so no sign error or missing term in the HJD_UTC $\rightarrow$ BJD_TDB conversion contributes. (iii) *The calibration sample*. Dropping WASP-18 b, the most precise of the four, moves the weighted mean to -0.22 ± 1.04 min (-0.72 ± 1.04 with the ExoClock ephemerides): **the bias rests on one calibrator at about 2.2σ** .

The bias is subtracted from every SuperWASP epoch and its uncertainty added in quadrature. The four rejected calibrators (between-season bars of 17–951 min) were not discarded

silently: a calibrator whose seasons disagree by tens of minutes carries no information on a 2-min bias, and the guard was a cap of 5 min on the between-season bar. That cap is a calibrator rule and is *not* applied to the targets — a target keeps whatever between-season bar its seasons give it, and a large one simply removes it from the curvature test (the two targets with $\varepsilon = 0$ in Sect. 5.4 have archival bars of 157 and 359 min).

What the bias does to the results. Its effect was measured rather than assumed, by refitting the 26 with the bias set to 0, -1.26 , -1.56 , -2.14 and -4.28 min and its uncertainty to 0.75 and 2.14 min (`src/sensib_vies_26.py`). The verdicts are nearly insensitive: D11 keeps all eleven members at the adopted value and at zero, and loses two only at -4.28 min or with the bias bar inflated to 2.14 min; TIC 198408416, 232634196, 329246824 and 392536812 — the four with the smallest relative uncertainty among the detections — survive every diagonal combination; the goodness-of-fit flags never change. The *values* are not insensitive: between a bias of 0 and -4.28 min the coefficient moves by more than its own σ in 16 of the 26 (median 1.36σ , maximum 4.70σ), so the bias is a first-order term in the tabulated dP/dt and a zeroth-order term in the verdicts. Choosing the ExoClock ephemerides instead (-1.56 min) changes no verdict at the adopted bias bar.

Declared limitation. The ± 0.75 min is a floor. The transfer of a bias measured on four WASP planets — the best-covered objects in the archive — to eclipsing binaries is asserted, not demonstrated, and the shape test above shows the estimator can bias a *given* target by a few minutes in either direction. The only eclipsing-binary check available is the data-level overlay of Sect. 5.5 (two SuperWASP-era epochs of CV Dra against independent CCD minima, differences -1.2 ± 2.7 and $+2.7 \pm 2.6$ min): consistent with the transferred bias, and far from a measurement of it. One more thing the ± 0.75 min is not: independent from one season to the next. Added in quadrature to each epoch, as the tables do, it averages down by $\sqrt{n}$ over a target's seasons, whereas the uncertainty of a chain bias is a common offset of the whole archival block — the shape most degenerate with curvature. The fits were therefore repeated by generalised least squares with that term moved from the diagonal into a common block over each target's SuperWASP epochs (`src/gls_vies_26.py`): no verdict changes in any of the 26, no new detection appears, and $\sigma(dP/dt)$ grows by a median of 2% and at most 16%. The tables keep the diagonal treatment; the correlated one is the correct propagation and changes nothing that is claimed. The residual test of Sect. 5.2.3 cannot bound any of this: a bias common to a target's seasons is a common offset of the archival block, which the quadratic absorbs, leaving no residual — an earlier version of this paragraph claimed the opposite (Appendix D).

3.5. Fit and tests

Linear and quadratic ephemerides are fitted by weighted least squares to all epochs (TESS sectors + SuperWASP seasons). $dP/dt = 2Q/P$ with Q the quadratic coefficient. Degrees of freedom are $N - 3$; a target with $N < 4$ would be “measured, not tested” (none of the 26 is). Curvature significance is $\Delta\chi^2$ between the two fits with one degree of freedom.

Three tests accompany every number. The *goodness of fit of the quadratic ephemeris* is the χ^2 of the fit with its $N - 3$ degrees of freedom, reported as p_{gof} ; a fit with $p_{\text{gof}} < 0.05$ is flagged “parabola inadequate” and its coefficient is not quoted as a rate. The level is the same as that of the curvature test, chosen for consistency and not for what it marks; an earlier draft

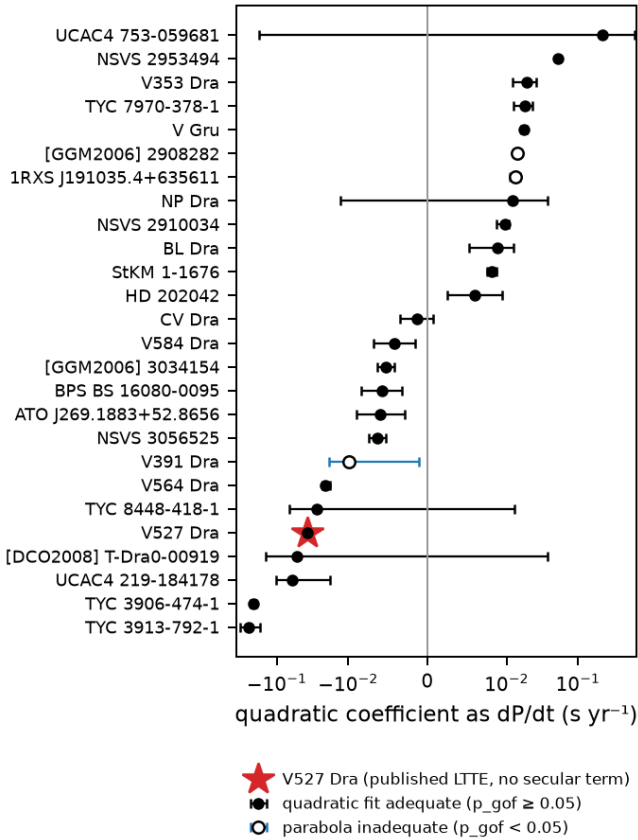


Fig. 1. Quadratic coefficients dP/dt of the 26 targets with 1σ bars, ordered by value; the three flagged targets ($p_{\text{gof}} < 0.05$) are shown open, and V527 Dra is marked.

used a fixed $\chi^2_{\text{red}} \geq 2$, which corresponds to a false-flag probability that depends on the degrees of freedom (0.157 at 1 d.o.f., 0.135 at 2, 0.112 at 3, 0.092 at 4) — the counts under both rules are reported in Sect. 4. The *curvature test* is the $\Delta\chi^2$ above. The *adversarial test* shifts every epoch by one sigma in the direction that most weakens the curvature and reports the resulting p-value; curvature is called only if $p < 0.05$ both before and after. The shifted value is not a p-value — a coherent one-sigma shift of every epoch has no null distribution — and the test is a robustness heuristic of unquantified conservatism, used here as a gate and not as a significance level. Sect. 5 shows what it does and does not protect against.

4. The chain applied to the 26

This section records what the chain returns with the data-derived bars of Sect. 3.3, before any of the tests of Sect. 5. Table 3 is the per-target record on which Sect. 5 operates; Table 4 summarises it in three partitions. Neither is a population result: the coefficients are quadratic coefficients over 16–20 yr, and Sect. 5 shows what they survive.

Five remarks on the table (Fig. 1 shows the coefficients; Fig. 3 shows what the detections track). (i) In every partition the distribution is consistent with symmetry about zero: 12/14 (sign test $p = 0.85$), 10/13 ($p = 0.68$) and 8/8 ($p = 1.00$). (ii) The median $|dP/dt|$ of 0.014 s yr^{-1} over all 26 is $1.6 \times 10^{-7} \text{ d yr}^{-1}$, the order of magnitude familiar from period studies of W UMa sys-

tems; 0.009 s yr^{-1} for the well-measured 16 is $1.0 \times 10^{-7} \text{ d yr}^{-1}$. That median is close to the amplitude at which the chain’s completeness for a secular term is 50% (0.014 s yr^{-1} ; Sect. 5.2.5). The 26 include the non-detections, so the distribution is not truncated, but it is shaped by the sensitivity of the chain at exactly the scale it reports, and the agreement with the W UMa literature — itself built from chains with their own thresholds — is not read here as an astrophysical result. (iii) Under the earlier fixed rule $\chi^2_{\text{red}} \geq 2$ the flag would have marked 5 targets instead of 3, and the chance that such a flag lands on one given target by coincidence is then 19% — a number that matters in Sect. 5.4. (iv) Five targets have $p_{\text{curv}} < 0.05$ and fail the adversarial gate (TIC 115244268, 199688409, 207496824, 229461186 and 359552377), and for them the gate, not a look-elsewhere correction, is what excludes them. (v) Two targets (TIC 224605072, 229476285) have no detection sensitivity at the amplitudes considered in Sect. 5.4 ($\varepsilon = 0$: their SuperWASP bars are too large for the curvature test to reject anything), so their “no significant curvature” is not a null result and they do not belong in the denominator of a curvature fraction; over the 24 with sensitivity the fraction is 11 of 24 (46%, 95% CI 28%–65%), given as a separate row of Table 4.

Two sets are used by name in the rest of the paper. **D11** is the eleven targets whose curvature survives the adversarial test (TIC 144194304, 198388252, 198408416, 229914020, 230386284, 232634196, 329246824, 377253090, 390021728, 392536812, 424461577); **D9** is D11 without the two flagged members (198388252, 229914020). One member of D9, TIC 377253090, has an unresolved companion of equal brightness at $1.7''$ (CROWDSAP 0.56; Sect. 3.1) and is carried with that caveat. The completeness of the chain for a secular term, measured by injection with the same estimator (Sect. 5.2.5), is 50% at 0.014 s yr^{-1} and, at the tabulated coefficients of the eleven, 0.65–1.00: the eleven are a count, not a rate. The third flagged target, TIC 237116051, has no surviving curvature. For the two flagged members of D11 the reading is *period variation detected without a determined form*; for the third, *not described by a quadratic ephemeris*; in neither case should the tabulated coefficient be quoted as a rate. Sect. 5.3.2 applies the measured noise model of the between-sector scale to this design and leaves 7 of the 11; Sect. 5.5 confronts the coefficients of D9 with the complete TESS block of the same targets. The sets have changed with the bars, and Appendix D keeps the record.

5. Three mechanisms that can produce a period change, and what the data can test

Each mechanism is stated, quantified on the 26, and confronted with what the data can and cannot test. The demonstration that frames all three is external and independent of the cache: for CV Dra, the one target with 34 years of independent CCD minima, a quadratic fitted to the same 2004–2024 window reproduces our coefficient (4 of 4 agree), and a quadratic fitted to the whole record does not (2 of 5; Fig. 2; Sect. 5.5). The object called “ dP/dt over 16–20 yr” is a local coefficient of a curve that is not polynomial; the three mechanisms below are the ways such a coefficient acquires a significance it does not have. The control target (TIC 142874476), on which the chain was developed, was re-run before and after every change reported in this version and reproduced each time; its record and the period-ladder audit are internal quality assurance and are given in Appendix B.

Table 3. Quadratic ephemerides of the 26 measured targets. Name is the variable-star designation where one exists, otherwise the SIMBAD identifier; positions are those of the TIC (J2000, degrees); T is the TESS magnitude; P the catalogue period in days; N the number of epochs; $\nu = N - 3$; span the baseline in years; dP/dr the quadratic coefficient expressed as a rate, in s yr^{-1} . Class: C, curvature surviving the adversarial test; –, none; a, fails the adversarial test; †, parabola inadequate ($P_{\text{gof}} < 0.05$), the coefficient then being that of a model that does not describe the data and listed for completeness only. No entry of this table is a period-change rate: each is a quadratic coefficient over 16–20 yr, and Sect. 5 shows what such a coefficient can be made of; the table is not to be cited as measured dP/dr (Sect. 6). V527 Dra (TIC 424461577) passes all three tests and has a published cyclic O–C (Sect. 5.4).

TIC	Name	RA	Dec	T	P	N	ν	span	dP/dr	χ^2_{red}	P_{gof}	P_{curv}	P_{adv}	cl.
115244268	TYC 7970-378-1	317.6428	-40.5239	10.9	2.0089	5 (3+2)	2	19.9	+0.0178 ± 0.0055	0.69	0.501	0.001	0.276	–a
126763885	UCAC4 219-184178	316.2145	-46.3262	12.4	4.3574	4 (2+2)	1	19.9	-0.0603 ± 0.0421	1.25	0.264	0.153	0.659	–
126945917	HD 202042	318.7324	-43.3954	9.1	1.3411	6 (4+2)	3	19.9	+0.0060 ± 0.0035	0.55	0.649	0.081	0.783	–
139256217	TYC 8448-418-1	347.1903	-46.1102	11.0	2.1992	4 (2+2)	1	19.9	-0.0274 ± 0.0403	1.10	0.295	0.496	0.859	–
144194304	V Gru	327.9727	-42.3732	9.2	0.4835	5 (2+3)	2	19.9	+0.0175 ± 0.0014	0.54	0.580	< 10 ⁻³	< 10 ⁻³	C
198388252	[GGM2006] 2908282	258.1953	+57.0529	11.6	0.3952	7 (4+3)	4	19.6	+0.0141 ± 0.0016	7.65	< 10 ⁻³	< 10 ⁻³	< 10 ⁻³	C†
198408416	NSVS 2910034	258.9133	+58.8645	12.1	0.3647	7 (4+3)	4	19.5	+0.0098 ± 0.0011	0.63	0.639	< 10 ⁻³	< 10 ⁻³	C
199688409	BPS BS 16080-0095	253.0515	+57.7255	12.0	0.8683	7 (4+3)	4	17.6	-0.0058 ± 0.0026	0.22	0.926	0.026	0.734	–a
207496824	V353 Dra	246.9546	+58.8398	10.8	0.8027	4 (2+2)	1	16.7	+0.0191 ± 0.0072	2.37	0.123	0.007	0.380	–a
224605072	[DCO2008] T-Dra0-00919	255.5992	+58.8700	11.4	0.6649	7 (4+3)	4	17.6	-0.0521 ± 0.0893	0.68	0.605	0.559	0.968	–
229461186	BL Dra	295.1026	+60.9203	10.9	0.4027	5 (3+2)	2	15.7	+0.0089 ± 0.0036	0.96	0.381	0.013	0.314	–a
229476285	UCAC4 753-059681	295.5345	+60.4830	12.9	1.0965	5 (3+2)	2	15.7	+0.2189 ± 0.3975	0.79	0.455	0.582	0.762	–
229687624	CV Dra	262.9885	+57.1784	9.0	0.6176	7 (4+3)	4	19.6	-0.0014 ± 0.0021	0.55	0.696	0.517	0.658	–
229914020	IRXS J191035.4+635611	287.6436	+63.9373	11.9	0.6760	5 (3+2)	2	15.7	+0.0132 ± 0.0023	5.42	0.004	< 10 ⁻³	< 10 ⁻³	C†
230386284	StKM 1-1676	285.8220	+63.9928	9.0	0.3415	5 (3+2)	2	15.7	+0.0081 ± 0.0006	0.07	0.929	< 10 ⁻³	< 10 ⁻³	C
232634196	TYC 3906-474-1	269.4385	+56.1960	9.9	1.2511	7 (4+3)	4	19.6	-0.2167 ± 0.0207	0.71	0.583	< 10 ⁻³	< 10 ⁻³	C
237116051	V391 Dra	269.7931	+58.7165	10.2	2.4629	4 (2+2)	1	19.6	-0.0098 ± 0.0087	13.25	< 10 ⁻³	0.260	0.589	–†
243352373	V584 Dra	290.2686	+56.3283	10.2	0.3343	4 (2+2)	1	16.7	-0.0041 ± 0.0026	0.75	0.388	0.117	0.855	–
329246824	V564 Dra	263.8847	+57.8025	11.0	0.5877	7 (4+3)	4	19.6	-0.0207 ± 0.0026	0.19	0.944	< 10 ⁻³	< 10 ⁻³	C
329248002	NP Dra	263.8179	+55.0034	8.8	3.1089	6 (4+2)	3	16.7	+0.0123 ± 0.0251	0.84	0.472	0.625	0.820	–
329289186	ATO J269.1883+52.8656	269.1883	+52.8657	13.8	0.5525	5 (2+3)	2	19.6	-0.0060 ± 0.0030	0.57	0.566	0.050	0.880	–
359552377	TYC 3913-792-1	279.3983	+57.7443	10.9	3.4391	5 (3+2)	2	16.7	-0.2498 ± 0.0765	0.68	0.508	0.001	0.212	–a
377253090	NSVS 3056525	286.1923	+58.9513	11.8	0.4245	6 (4+2)	3	16.7	-0.0063 ± 0.0010	0.51	0.673	< 10 ⁻³	< 10 ⁻³	C
390021728	[GGM2006] 3034154	277.7934	+56.4175	11.5	0.3020	6 (4+2)	3	16.6	-0.0053 ± 0.0010	1.15	0.326	< 10 ⁻³	0.003	C
392536812	NSVS 2953494	280.4674	+58.0565	11.2	0.5910	6 (4+2)	3	16.7	+0.0532 ± 0.0029	0.24	0.871	< 10 ⁻³	< 10 ⁻³	C
424461577	V527 Dra	279.7343	+60.3982	10.2	0.7448	6 (4+2)	3	15.7	-0.0370 ± 0.0033	2.04	0.106	< 10 ⁻³	< 10 ⁻³	C

Table 4. The distribution, in three partitions: all 26 coefficients; the 23 whose quadratic fit is adequate ($p_{\text{gof}} \geq 0.05$); and the 16 that are adequate and have $\sigma(dP/dt) < 0.01 \text{ s yr}^{-1}$, a cut on precision rather than on value. No column is a headline: the median $|dP/dt|$ moves from 0.009 to 0.018 s yr^{-1} between them, and that sensitivity is the finding.

Quantity	n = 26 (all)	n = 23 (quadratic fit adequate, $p_{\text{gof}} \geq 0.05$)	n = 16 (adequate and $\sigma(dP/dt) < 0.01 \text{ s yr}^{-1}$)
dP/dt , median	-0.0027 s yr^{-1}	-0.0041 s yr^{-1}	+0.0023 s yr^{-1}
$ dP/dt $, median	0.0137 s yr^{-1}	0.0175 s yr^{-1}	0.0085 s yr^{-1}
dP/dt , interquartile range	[-0.0180, +0.0130] s yr^{-1}	[-0.0241, +0.0111] s yr^{-1}	[-0.0058, +0.0118] s yr^{-1}
dP/dt , minimum and maximum	[-0.250, +0.219] s yr^{-1}	[-0.250, +0.219] s yr^{-1}	[-0.037, +0.053] s yr^{-1}
Sign of dP/dt : positive / negative (two-sided sign test p)	12 / 14 (p = 0.85)	10 / 13 (p = 0.68)	8 / 8 (p = 1.00)
$\sigma(dP/dt)$, median	0.0032 s yr^{-1}	0.0033 s yr^{-1}	0.0026 s yr^{-1}
χ^2_{red} of quadratic fit, median (25th–75th)	0.70 (0.55–1.14)	0.68 (0.55–0.90)	0.56 (0.44–0.80)
$ dP/dt /\sigma > 2 / > 3 / > 5$	16 / 13 / 11	14 / 11 / 9	12 / 9 / 8
Curvature with p < 0.05	16	14	12
Curvature surviving the adversarial test	11 (42%, 95% CI 26%–61%) — 6 > 0, 5 < 0	9 (39%, 95% CI 22%–59%) — 4 > 0, 5 < 0	8 (50%, 95% CI 28%–72%) — 4 > 0, 4 < 0
Same, over targets with detection efficiency $\varepsilon > 0$ (Sect. 5.4)	11 of 24 (46%, 95% CI 28%–65%)	9 of 21 (43%, 95% CI 24%–63%)	—
Same, with TESS bars inflated by 1.0 / 2.6 min yr^{-1} (Sect. 5.3, sensitivity ceiling)	1 / 0 of 26	1 / 0 of 23	0 / 0 of 16
Measured but not tested (d.o.f. < 1)	0	0	0
$ dP/dt > 10 \text{ s yr}^{-1}$ (cycle-count guard)	0	0	0

Table 5. The window test and the sensitivity to the archival bias, per target. For the nine members of D9: the quadratic coefficient of the complete TESS block with the four design epochs removed (pooled-GP fit, the variant comparable across targets; number of epochs in brackets), the range of z against the tabulated coefficient over the noise models admitted for that target (Sect. 5.3.2), the resulting class (Sect. 5.5), and MDD, the smallest difference the block could have detected at 3σ under the admitted model. V527 Dra is held out of the classification and carries its published light-travel-time orbit instead; its complete fit including the archival seasons is compatible with zero (Sect. 5.4). The last column, given for all 26, is the change of the tabulated coefficient of Table 3 per 2.14 min of archival bias (Sect. 3.4) — the bias is a first-order term in the value and a zeroth-order term in the verdict.

TIC	Name	complete block (s yr^{-1})	N	z (admitted)	class	MDD	Δ per 2.14 min
115244268	TYC 7970-378-1	—	—	—	not in D9	—	0.0053
126763885	UCAC4 219-184178	—	—	—	not in D9	—	0.0117
126945917	HD 202042	—	—	—	not in D9	—	0.0036
139256217	TYC 8448-418-1	—	—	—	not in D9	—	0.0059
144194304	V Gru	—	4	—	not testable	—	0.0013
198388252	[GGM2006] 2908282	—	—	—	not in D9	—	0.0011
198408416	NSVS 2910034	+0.0453 $\pm$ 0.0121	35	+3.4 to +6.1	contradicted	0.0321	0.0011
199688409	BPS BS 16080-0095	—	—	—	not in D9	—	0.0030
207496824	V353 Dra	—	—	—	not in D9	—	0.0028
224605072	[DCO2008] T-Dra0-00919	—	—	—	not in D9	—	0.0020
229461186	BL Dra	—	—	—	not in D9	—	0.0016
229476285	UCAC4 753-059681	—	—	—	not in D9	—	0.0026
229687624	CV Dra	—	—	—	not in D9	—	0.0017
229914020	1RXS J191035.4+635611	—	—	—	not in D9	—	0.0026
230386284	StKM 1-1676	+0.0088 $\pm$ 0.0110	35	+1.0	not contradicted	0.0024	0.0013
232634196	TYC 3906-474-1	-0.7005 $\pm$ 0.0476	39	-9.5 to -3.2	contradicted	0.3983	0.0036
237116051	V391 Dra	—	—	—	not in D9	—	0.0057
243352373	V584 Dra	—	—	—	not in D9	—	0.0012
329246824	V564 Dra	-0.0036 $\pm$ 0.0197	38	+2.9	ambiguous	0.0197	0.0017
329248002	NP Dra	—	—	—	not in D9	—	0.0111
329289186	ATO J269.1883+52.8656	—	—	—	not in D9	—	0.0016
359552377	TYC 3913-792-1	—	—	—	not in D9	—	0.0123
377253090	NSVS 3056525	-0.0027 $\pm$ 0.0140	36	+0.0 to +0.3	not contradicted	0.0146	0.0015
390021728	[GGM2006] 3034154	+0.0269 $\pm$ 0.0103	37	+3.8 to +5.5	contradicted	0.0263	0.0011
392536812	NSVS 2953494	+0.0181 $\pm$ 0.0217	38	-4.0	contradicted	0.0223	0.0021
424461577	V527 Dra	+0.0508 $\pm$ 0.0265	36	—	published LTTE	—	0.0029

5.1. Formal timing errors

Formal errors against the residuals. The reduced χ^2 of the quadratic fit is the check that the assigned errors describe the residuals. With formal errors it had median 6.35 (25th–75th: 2.08–37.5), was below 2 for only 8 of 31, and left maximum quadratic residuals of 4.4 min (median) to 18 min (75th percentile) against bars of 0.4–0.7 min: not even the more complex model explained the residuals, which locates the problem in the denominator, not the model.

Consequence of the formal-error run. That run — kept in full as `oc_lote_barras_formais/` — returned 24 of 31 tar-

gets with $|dP/dt|/\sigma > 3$ and 23 surviving the adversarial test; with the data-derived errors the count is 11 of 26. The 13-sigma “detection” that headed the formal-error list (TIC 199688472, +1.75 $\pm$ 0.006 s yr^{-1}) was a quadratic bending to reach two SuperWASP seasons of ± 76 min at 19 yr of leverage, one of which had a negative fitted depth; the target is now in Table 2, and the entire positive tail of that distribution (95th percentile +0.84 s yr^{-1} , now a maximum of +0.22) consisted of such targets. Formal timing errors in archival photometry produce spurious period changes in bulk, and they produce them preferentially with large leverage, where they look most significant.

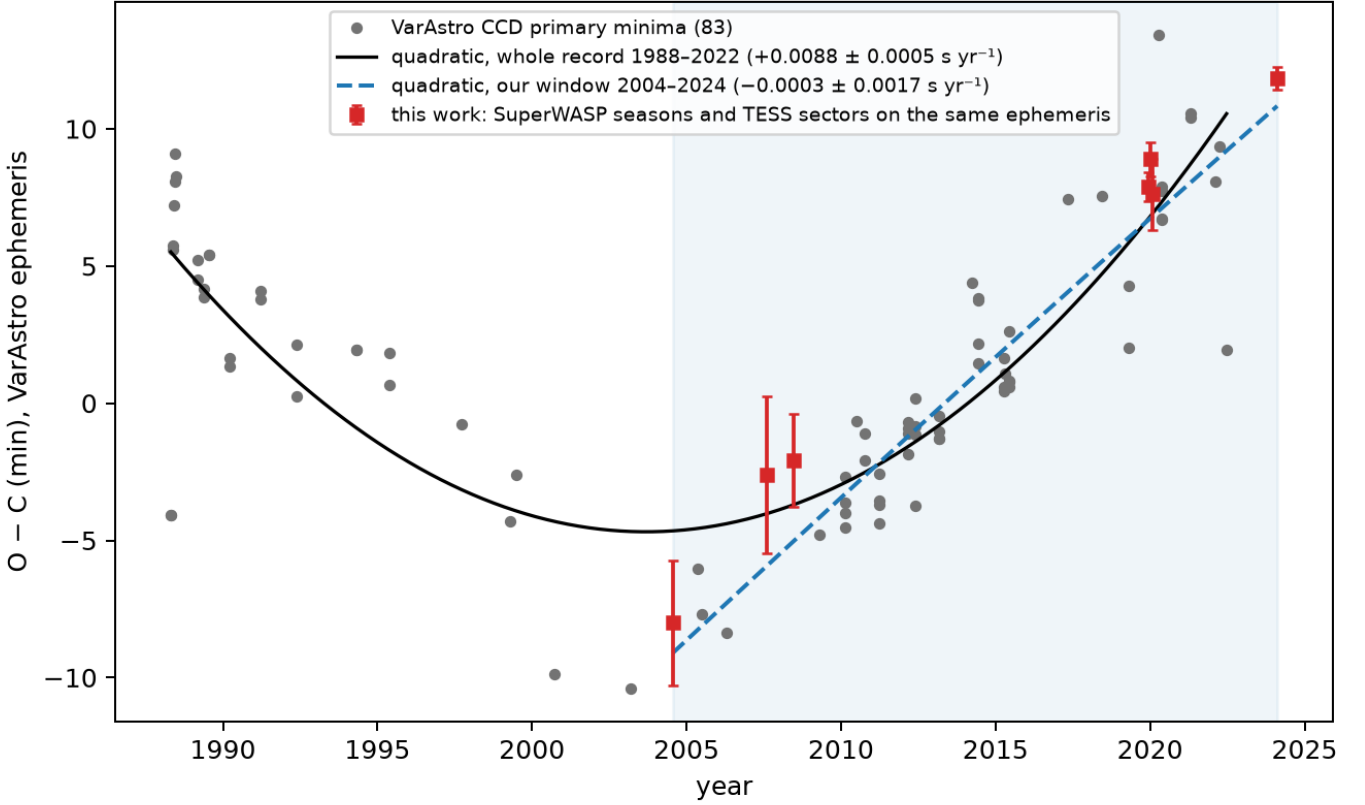


Fig. 2. CV Dra: VarAstro CCD primary minima 1988–2022 with the quadratic fitted to the whole record and the one fitted to our 2004–2024 window, and our epochs placed on the same ephemeris.

5.2. The residual test and its null: what data-derived bars can be validated against

This subsection is about the test that validates data-derived bars — the residuals of the quadratic fits, normalised by the bar of each epoch, corrected for leverage and summed by data set — and its result is a property of the test rather than of the sample. Stated first: when each bar is estimated from the same two or three samples that enter the residual, the null of the test is **0.77 and not 1**, so a reduced χ^2 near 0.7 is what correct bars produce in this design. Against that null the bars of Sect. 3.3 are consistent with the residuals of the 26 except in three targets whose quadratic the goodness-of-fit test rejects, which hold the excess of both archives; over the other 23 both archives are inside their nulls once the same cut is applied inside the simulation. The reading is narrower than “the bars are right”, because the test cannot separate correct bars from bars inflated by real secular terms of the tabulated size. The path to this result passed through an archival excess of 2.73 that three tests treated as real — a composition error of 1.74 in the bars, and the remainder measured against a null of 1 — and the readings built on it are withdrawn (Appendix D); the simulations and their committed expectations are in Appendix E.

5.2.1. The data-derived bars against the expected reduced χ^2

With errors from the data (Sect. 3.3) the median reduced χ^2 of the 26 quadratic fits is 0.70. With 1–4 degrees of freedom the median of χ^2/ν under correct bars is well below 1, and for the actual distribution of degrees of freedom of the 26 the expected

median is 0.73 (95% range 0.43–1.15; 20 000 simulations); 0.70 has probability 0.57 of being this high. The aggregate does not: reduced χ^2 1.44 over 67 degrees of freedom, of which the three targets the goodness-of-fit test rejects contribute more than half, the other 23 giving 0.71. Both comparisons assume the bars are known; they are not, and Sect. 5.2.3 simulates the null of the pipeline’s own estimator. Against that null the median reduced χ^2 is 0.74 (2.5–97.5%: 0.54–0.95) and the measured 0.70 sits at its 38th percentile; the aggregate null is 0.90 (0.65–1.26) and the measured 1.44 is above every one of its 2 000 realisations — an excess, which the decomposition of the next two subsections locates in the three flagged targets: without them the aggregate of the 23 is inside the null. Earlier versions measured different values here and built a bar correction on them; the difference is the composition error of the next subsection, and those readings are withdrawn (Appendix D).

5.2.2. The leverage-corrected decomposition by data set, and the composition error it exposed

A global χ^2 says whether the total variance is described, not where a problem sits. The residuals of the quadratic fits, normalised by the bar of each epoch and corrected for leverage — with three parameters on 4–7 points, $E[r^2/\sigma^2] = 1 - h$, and h reaches 0.6 for the TESS epochs — were summed separately over the 84 TESS and 61 SuperWASP epochs of the 26 targets (src/qual_barra_26.py; expectation written before running: excess in TESS). The SuperWASP epochs give $\Sigma z^2 = 46.5$ against 35.4 expected under known bars, a variance factor of 1.31 (χ^2 interval 0.87–2.23; bootstrap over targets 0.62–2.30);

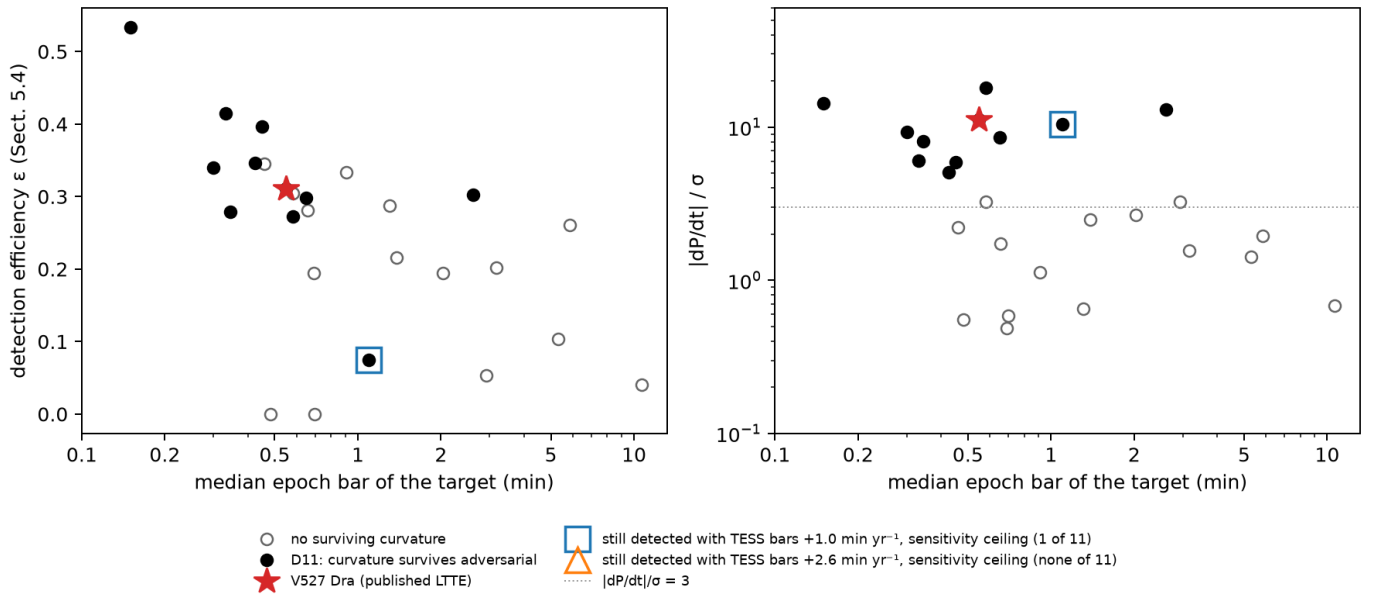


Fig. 3. Detection efficiency, significance and the bars. Left: the per-target detection efficiency ε of Sect. 5.4 against the median epoch bar of the target; right: $|dP/dt|/\sigma$ against the same bar. Filled symbols are D11 and the star is V527 Dra; the open squares and triangles mark the targets still detected under the *ceiling* of Sect. 5.3.2, with the TESS bars inflated by 1.0 and 2.6 min yr⁻¹ (one target and none) — not under the admitted noise model of record, which leaves seven (Table 5 gives the window test of each). The dotted line in the right panel is $|dP/dt|/\sigma = 3$.

the TESS epochs give 50.2 against 31.6, a factor of 1.59 (1.02–2.79; 0.54–3.08). The mean leverage is 0.62 for the TESS epochs and 0.42 for the SuperWASP ones: the test retains 38% of the TESS variance and 58% of the SuperWASP variance. Under the TESS bar of the previous version (Sect. 3.3) the TESS factor was 1.02 (0.66–1.79): the $\sqrt{2}$ in the bar hid a factor of 1.5 in the TESS residuals, and where that factor sits is located below.

These are not the numbers the first decomposition returned: it gave a SuperWASP factor of 2.73, read as bars 1.65 times too small, because the routine that measures the between-season dispersion returned the standard error of the season *mean* and the batch chain needs the dispersion itself — a factor of 1.74 in variance, at the seam between two callers of one function, where the control target could not see it. The correction, what it moved and the version built on the earlier number are in Appendix D.

The test is asymmetric: it has power for SuperWASP, whose epochs sit at the end of the lever, and little for TESS, whose four epochs are fitted by three parameters — what the parabola absorbs does not appear as a residual. A TESS excess at the between-sector scale is therefore not excluded by this test, only not required by it. The estimator itself is unbiased when the bars are known, and a coherent structure at the TESS epochs raises the TESS factor first — which is how the measured 1.59 bounds such structure at about 2.5 min (Appendix E). What the test cannot see is structure *inside* the parabola — a drift linear or quadratic over four epochs is absorbed into the coefficient — and that is the blindness that matters, because it is exactly the between-sector drift of Sect. 5.3.

5.2.3. The null of the estimator: bars re-estimated from two or three samples

The χ^2 interval quoted above assumes the bars are known. They are not: each SuperWASP bar is the standard deviation of 2–3 season deviations that also enter the residual, and each TESS bar is $\max(\text{formal}, |t_a - t_b|/2)$ from the two halves of the sector — in both cases s is estimated with $\nu = 1$ –2 degrees of freedom

from the same points as r , so $z^2 = r^2/s^2$ has no χ^2 distribution and the null has to be simulated as the pipeline operates: the 26 real designs, Gaussian noise with true σ equal to the assigned bar and no curvature, the bars re-estimated from the simulated seasons and sector halves by the pipeline’s own rule, 2 000 realisations (`src/nulo_barras_reestimadas_26.py`; the expectation written before it, and the argument that motivated it, are in Appendix E).

The expectation was wrong in direction. With bars re-estimated, the null of the SuperWASP factor is 0.77 (2.5–97.5%: 0.63–0.94; quartiles 0.72–0.81), not above 1, and the TESS null is 0.98 (median; mean 1.03; 0.54–1.79). The $\nu/(\nu - 2)$ argument assumes r independent of s ; here they are the same numbers. What deflates the null is the floors: the bar is the larger of the co-estimated dispersion and the formal error and carries the bias term in quadrature, so whenever the dispersion of two or three samples falls below a floor the bar exceeds it while the residual does not. Removing the floors in the simulation (`src/nulo_pisos_26.py`) returns the SuperWASP null to 1.15 and makes the TESS null diverge, because a half-sector dispersion near zero makes z unbounded; all three floors together give 0.78 and 0.95. The floors are what make the statistic usable, and also what set its null below 1. With the bars held fixed the same simulation returns 1.00 in both sets: the estimator is unbiased when σ is known and biased low when σ is co-estimated, and the second case is the pipeline. A reduced χ^2 near 0.7 is therefore what correct bars *produce* in this design; it is the number to compare with, not a sign of over-estimated bars. The value is a property of this design and not a constant — another sample, another number of seasons per target or another floor gives another null, and what transfers is the simulation, not the number (Appendix E).

Against its own null the measured SuperWASP factor of 1.31 is not a small- ν artefact but an excess: none of the 2 000 realisations reaches it. The measured TESS factor of 1.59 sits at the 94th percentile of its null — a milder excess, on the side the test has least power for. Where it sits is a reading made after the fact and labelled as such, but the criterion it uses is the one Sect. 3.5

declares, not one chosen after seeing the residuals: the three targets whose quadratic the goodness-of-fit test rejects ($p_{\text{gof}} < 0.05$; TIC 198388252, 229914020, 237116051) contribute more than half of the 46.5 on the SuperWASP side, and 19.6 (198388252) and 10.7 (229914020) of the 50.2 on the TESS side. Over the 23 targets with an adequate quadratic the SuperWASP factor is 0.72 and the TESS factor 0.68, each at the 24th percentile of a null simulated with the *same* goodness-of-fit cut applied inside every realisation — the conditioning matters, and how it was done is in Appendix E. The excess is not in the bars of either archive; it is in three targets whose parabola does not describe the data, and one of them, TIC 198388252, is a member of D11 — a detection that passes the significance and adversarial tests on a quadratic the goodness-of-fit test rejects, the same structure as V527 Dra with a different confounder (Sect. 5.4). No bar correction is applied: the bars of Sect. 3.3, with the composition error corrected, are the bars of Tables 3–5, and the earlier “corrected” and “jointly corrected” sets of detections are withdrawn (Appendix D).

What the test cannot separate. A real secular term inflates the co-estimated archival bar (Sect. 5.2.5), and an inflated bar lowers the factor, so a sample whose bars are right and a sample carrying real terms of the tabulated size give the same reading. Injecting a term in every target and re-measuring (`src/completeness_secular_26.py` -fator) makes the archival factor fall monotonically with the amplitude while the TESS factor stays put, and the measured 0.72 of the 23 is as likely under no term as under $0.01\text{--}0.02\text{ s yr}^{-1}$ in every target; what the test does exclude is a term of 0.05 s yr^{-1} in every target. The mechanism is the coupling between bar and residual, not a migration of the fit to the TESS block — the leverage of the archival epochs does not move — and the grids of both are in Appendix E. **What this subsection can say is therefore narrower than “the bars are right”: the bars are consistent with the residuals, and so would be bars inflated by real terms of up to about 0.02 s yr^{-1} in every target.**

5.2.4. What the residuals leave for orbits, with the same estimator

An undersampled third-body orbit is the structure that the four TESS epochs absorb into the parabola and the SuperWASP seasons, 12–16 yr away, do not. How much of an archival excess such orbits could produce was measured by injecting the declared population of Sect. 5.4 into the 26 designs with occurrence f , with the bars re-estimated inside the simulation, and reading the two factors (`src/estructura_swasp_26.py`; the first run held the bars fixed, whose null is 1.00 against the measurement’s 0.77, and the numbers it gave are withdrawn as cross-estimator comparisons — Appendix D). Two results, with the grid in Appendix E: the re-estimated bars absorb most of an orbit’s between-season signal, so **the archival side is nearly blind to orbits** and the TESS side is where they would show; and the archival excess of the three flagged targets is not an orbit population, since even at an occurrence of one the probability that orbits alone produce the measured SuperWASP factor is 0.16. The TESS side places no useful bound on the tertiary fraction: nothing below the highest published value is excluded, and the central value is taken from Tokovinin et al. and not from here.

5.2.5. A secular term through the same estimator: completeness and self-suppression

The two subsections above ask what the estimator does to noise and to orbits; this one asks what it does to the signal the paper is nominally about. A pure secular term was injected on the 26 real designs at $|dP/dt| = 0.005\text{--}0.25\text{ s yr}^{-1}$ (the median tabulated $|dP/dt|$ is 0.014), with the bars re-estimated inside each realisation by the pipeline’s rule — for SuperWASP the dispersion of the simulated seasons *with the signal in them*, because a real secular term appears as season-to-season spread — and each realisation judged by the curvature test and the adversarial gate (`src/completeness_secular_26.py`; the full setup and the two committed expectations are in Appendix E). Fig. 4 shows the two curves of every target.

Completeness. With no term injected the gate fires in 0.0% of realisations at the median target and in at most 6% in 23 of the 26; summed over the 26 the false-alarm rates give 0.39 detections expected from noise alone, and no member of D11 has a rate above 1.0%. The amplitude at which half the injected terms are detected has a median of 0.014 s yr^{-1} — the median tabulated coefficient is 0.014 — and completeness at 0.02 s yr^{-1} is ≥ 0.5 in 16 of the 26; at 0.2 s yr^{-1} it exceeds 0.9 in 21 and stays below 0.75 in the three targets with archival bars above 15 min, which are also the three that never reach 0.5 on the grid. At the coefficient each of them carries in Table 3, the eleven members of D11 have a completeness of 0.65–1.00 (median 0.94): the denominator of “eleven detections” is not eleven, and no count of detections in this paper is a rate. What sets the 50% amplitude, and what re-estimating the bars costs in completeness, are in Appendix E.

Self-suppression. The re-estimated bars contain the signal, and the signal inflates them. Against the expectation of a 20–40% effect at 0.2 s yr^{-1} , $\sigma(dP/dt)$ with re-estimated bars exceeds σ with the true bars by a median factor of 1.03 at 0.02 s yr^{-1} , 1.19 at 0.05, 1.6 at 0.1 and 2.9 at 0.2 — and by 5–24 at 0.2 in the 9 targets whose archival seasons have bars of 1.5–5 min at the longest lever (up to $\times 24$). The simulated noise is the assigned bar of each epoch, which on a target with real structure already contains it, so the suppression measured is a floor. The reason for it is arithmetic: a term of 0.02 s yr^{-1} at $E \approx -8000$ cycles moves two seasons 730 cycles apart by about 3 min relative to each other, the size of the bars; at 0.2 s yr^{-1} the between-season spread of the signal is tens of minutes and the “bar” is made of it. The coefficient itself is unbiased (median recovered/injected 1.00–1.01) and the goodness-of-fit cut does not see the term: the estimator cannot manufacture a detection from a secular term, but it turns the tabulated σ of a real one into an overestimate that grows with the term, and it hides real terms near the 50% point. This is an estimator property of the same family as the composition error of Sect. 5.2.2 — each season’s bar estimated from the seasons themselves — and, unlike it, not corrected here; the alternatives that would break the coupling, none implemented in this version, are listed in Appendix E. For TIC 232634196 the simulation on its own design, with the noise taken as its 19.9-min bars, gives a completeness of 0.94 and a suppression of 1.7 at its coefficient; if those bars were instead the signal’s own spread on the 1.4-min formal noise — which is what the simulation produces for a term of that size on the small-bar designs — the tabulated σ would be an overestimate by an order of magnitude. The data decide against the second reading on one point: if the 19.9 min were the term’s own spread on 1.4-min formal noise, the three seasons would fall on the fitted parabola within about 1.4 min, and they fall at 13, 0 and -22 min from it.

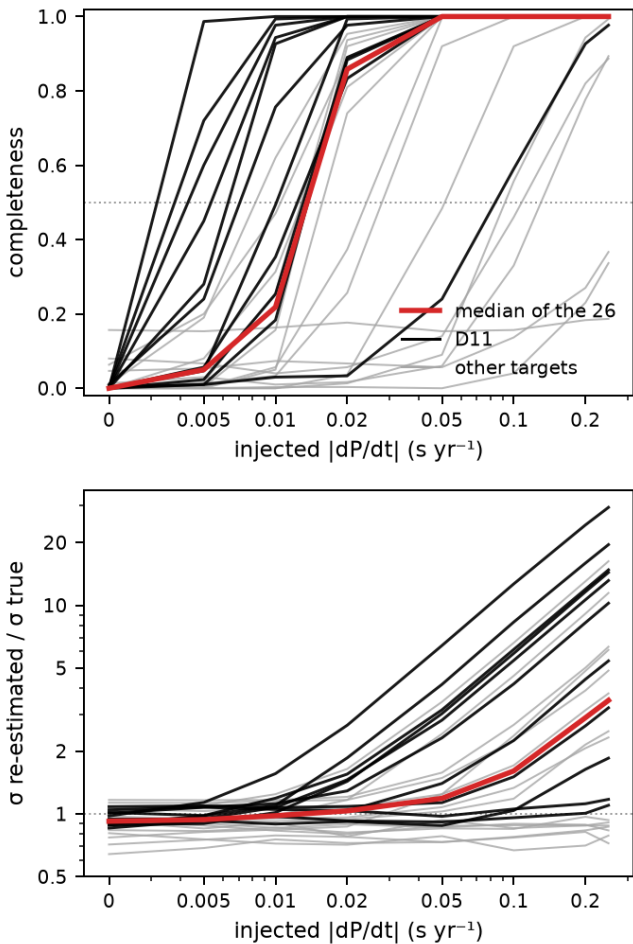


Fig. 4. A secular term through the estimator of the chain (Sect. 5.2.5). Top: completeness — the fraction of injected terms passing the curvature test and the adversarial gate with the bars re-estimated inside the simulation — against the injected $|dP/dt|$, one curve per target (black: D11; grey: the others; red: the median of the 26; the point at 0 is the false-alarm rate). Bottom: the suppression factor, $\sigma(dP/dt)$ with the re-estimated bars over σ with the true bars, on the same grid. Both are properties of the 4–7-epoch design, not of the complete TESS block of Sect. 5.5.

5.3. Between-sector timing noise: the secondaries and the TESS sensitivity ceiling

5.3.1. Secondary minima: the differential drift between sectors

Apsidal motion in an eccentric orbit produces a spurious quadratic in the primary alone, with the secondary moving the opposite way; a secular dP/dt moves both together. The secondary epoch was measured in every TESS sector of the 26 with the same estimator, the catalogue seed shifted by $P/2$, the same trapezoid and the same halves-dispersion bar (`src/secondarios_26.py`; expectation written before running: most secondaries measurable, 0–2 targets with a significant differential drift, up to 5 with no measurable secondary). The quantity tested is $d = t_{\text{sec}} - t_{\text{prim}} - P/2$ per sector, and its drift between the first and last sector (median lever 4.2 yr within TESS).

All 26 secondaries were measurable; for two of the long-period targets (TIC 139256217, 359552377) the bar on d exceeds 10 min and the test is uninformative, leaving 24. In 7 of the 24 the

secondary sits more than 2σ from phase 0.5 (TIC 126763885, 198388252, 224605072, 229914020, 232634196, 329246824, 329289186), which is a statement about eccentricity or asymmetry, not about period change. In 6 of the 24 the differential drift is significant at $> 2\sigma$ — more than the 0–2 expected: TIC 115244268 ($+1.26 \pm 0.50 \text{ min yr}^{-1}$), 229476285 (-1.50 ± 0.31), 229914020 ($+2.59 \pm 0.25$; flagged), 232634196 (-1.80 ± 0.49), 329289186 ($+2.41 \pm 0.42$), 390021728 ($+0.67 \pm 0.24$) (under the TESS bar of the previous version the sixth, TIC 115244268, was at 1.9σ and five were significant). One of the four targets with $P > 2 \text{ d}$ and an informative secondary shows a drift (TIC 115244268, $P = 2.01 \text{ d}$, at 2.5σ), so the eccentric-orbit confounder is not excluded where it was expected; the other five drifts occur in systems with $P = 0.3\text{--}1.25 \text{ d}$, where migrating spots are the usual cause of differential primary/secondary timing.

Of the eleven detections of D11, three are affected. TIC 229914020 has the largest differential drift of the sample ($+2.59 \pm 0.25 \text{ min yr}^{-1}$, 10σ), is flagged by the goodness-of-fit test, and is in D11: its curvature, its bad fit and its drifting secondary are three views of the same non-quadratic timing behaviour, and the target belongs to none of the categories that read as period change. TIC 232634196 has a differential drift of $-1.8 \pm 0.5 \text{ min yr}^{-1}$ over 4.2 yr, which extrapolated over the 20-yr baseline — an illustration, since it was measured on a fifth of it — would be 36 min against a quadratic excursion of 201 min, so the secondary shares part of that curvature and not necessarily all of it. TIC 390021728 has the smallest of the six drifts ($+0.67 \pm 0.24 \text{ min yr}^{-1}$), which extrapolated the same way is 13 min against a quadratic excursion of 13 min: its curvature and its secondary's drift are of the same order. The primary/secondary test qualifies three of the eleven; it does not bear on the third-body mechanism of Sect. 5.4, which moves both minima together.

The excess itself — six drifts against 0–2 expected — is the measurement that motivates the next subsection: a drift of $0.7\text{--}2.6 \text{ min yr}^{-1}$ between sectors years apart is invisible to a within-sector bar and to the residual test. Whether the primaries drift as the secondaries do is no longer left open here: the 799 cached sectors measure it directly.

5.3.2. The noise model at the between-sector scale, and what it does to the design

The secondary-minimum test above finds the secondary drifting with respect to the primary by $0.7\text{--}2.6 \text{ min yr}^{-1}$ between sectors in six of 24 targets. The TESS bar of Sect. 3.3 is the dispersion between the two halves of a sector, a within-sector quantity over $\sim 13 \text{ d}$, and cannot contain such a drift; the residual test of Sect. 5.2 cannot see it either, because a drift that is linear or quadratic over four epochs is absorbed by the fit. The question is therefore not whether to test the primary epochs at that scale but with what noise model, and the 799 cached 2-min light curves of the 26 make the model measurable instead of assumed.

The measurement. Every primary epoch of every cached sector was measured with the estimator of Sect. 3.1 (799 sectors; the four epochs of each design are reproduced to 10^{-3} min , which is the control of this step). Two descriptions were fitted to each target's extended block: a white excess σ_{w} added in quadrature to the bars, by maximum likelihood against a local quadratic ephemeris, and a stationary Gaussian process with a Matérn-3/2 kernel of amplitude A and correlation time τ_{c} . The white excess has a median of 0.66 min against a linear ephemeris and 0.42 min against a quadratic — the difference is real curvature absorbed by the linear model — and is required at 95%

in 17 of the 21 targets where the block is informative; the declared stopping rule (σ_j above 3 min, which would have meant the epochs were not measuring the same event) never fires. The between-sector noise is not white: a variogram of the residuals rises from $\gamma \approx 1.3$ below 30 d to a plateau at 1–4 yr, and a linear fit gives $q = 0.257 \text{ min}^2 \text{ yr}^{-1}$ with a 95% interval of -0.019 to 0.622 , so a random walk is not required at 2σ and the process is treated as stationary. Pooled over the 15 targets outside D11, the Matérn-3/2 kernel gives $A = 0.835 \text{ min}$ and $\tau_c = 68 \text{ d}$ (the exponential kernel gives 0.824 min and 91 d and is disfavoured by $\Delta(-2 \ln L) = 18$).

Which noise model is admitted. The pooled amplitude is not the right one for every target: fitted per target, A ranges from 0 (six targets, among them TIC 230386284 and 392536812) to 5.5 min (V527 Dra, where it is fitting the published light-travel-time orbit). A model is *admitted* here if its reduced χ^2 on the extended block falls inside the 95% interval of χ^2_{ν}/ν for that target's ν — a criterion that uses goodness of fit only and never the coefficient. Of the two candidate models with a per-target parameter, the white σ_j is admitted in 24 of the 26 targets and the per-target GP in 22, both in 22; two targets (TIC 230386284 and 329248002) admit neither, their extended blocks giving $\chi^2_{\nu}/\nu = 0.17$ – 0.28 , i.e. per-sector bars that are conservative rather than optimistic, and for them the least rejected model is used and the line is marked. Where both are admitted the one with χ^2_{ν}/ν closer to 1 is applied, which gives the GP in 15 of the 26 and the white model in 11 (the two marked targets among them). The criterion was written after the comparison of the noise models, not before, and uses χ^2_{ν}/ν alone; that order is stated because it matters for how the result is read.

Applied to the design, 7 of the 11 detections survive (TIC 144194304, 198408416, 230386284, 232634196, 329246824, 377253090 and 392536812; the four that fall are 198388252, 229914020, 390021728 and 424461577), no new detection appears, and the count is unchanged if the admission interval is widened to 99% or narrowed to 90%. Three of these seven are contradicted by the complete TESS block of their own target (Sect. 5.5), so surviving the noise model is not the same as surviving the data. This is the noise model of record for the design; two sensitivity variants are reported beside it. The first is white inflation by the secondaries' drift (last row of Table 4): at 1.0 min yr^{-1} one detection survives and at 2.6 none, an extrapolation of a drift measured on the secondary to the primary, conservative by construction — it adds $d \times L$ in quadrature to every TESS epoch independently, modelling a time-correlated drift as white noise of 4–11 min — and a ceiling that the TESS residuals neither require nor exclude. The second is the *pooled* GP, which also leaves 7 of 11 but not the same seven (198388252 and 424461577 in, 230386284 and 377253090 out): **the pooled amplitude is not calibrated in any of the eight targets where the complete block can test it** (χ^2_{ν}/ν from **0.05** to **6.4**), over-covering the quiet targets and under-covering the noisy ones, so it is a comparability device and not a description.

What an amplitude does to a coefficient. The effect is large and it is worth stating in the open, because it is the reason the noise model has to be chosen by a declared criterion. For TIC 230386284, whose 38 sectors have 0.10-min bars, $\sigma(dP/dt)$ scales monotonically and almost linearly with A once A exceeds the per-epoch bar: $0.00081 \text{ s yr}^{-1}$ at $A = 0$, 0.00279 at 0.2 min , 0.00534 at 0.4 , 0.01098 at the pooled 0.835 , 0.02615 at 2.0 , while the coefficient itself moves by less than 3% ($+0.0091$ to $+0.0083$). The ratio between the pooled and the diagonal σ runs from 2.3 to 13.6 across the eight targets and tracks A divided by the per-sector bar (Spearman $+0.93$, $p = 0.001$). The mechanism

is the correlation time: with $\tau_c = 68 \text{ d}$ the 38–43 sectors of a target fall into four observing blocks separated by more than τ_c (three by more than $3\tau_c$, and the counts are the same for all eight), and within a block the process imposes what is nearly a common offset, so the quadratic rests on four nearly-independent offsets rather than on 38 epochs. An amplitude taken from other targets therefore sets the error bar of this test, which is why it is admitted or rejected by the data of the target itself.

5.4. Undersampled third-body orbits

V527 Dra: a false positive of secular curvature that passes every internal test. The eclipse-timing analysis of V527 Dra (Čeki et al. 2024; 1 832 timings including 34 from SuperWASP) is a light-travel-time orbit with $P_3 = 2.733 \pm 0.005 \text{ yr}$ and semi-amplitude 6.6 min , *with no secular term*; the star is also one of the eleven objects SIMBAD links to the TESS full-frame-image timing study of Marcadon & Prša (2024), which reports ten new triple candidates. Our chain, with six epochs and no access to any of those timings, returns $dP/dt = -0.0370 \pm 0.0033 \text{ s yr}^{-1}$ with $p_{\text{curv}} \approx 10^{-27}$, $p_{\text{adv}} \approx 10^{-17}$, $p_{\text{gof}} = 0.11$: a curvature detection that survives the adversarial test and whose quadratic fit is not rejected, in a system whose published O–C has no secular term. **The one target with an external record of a cyclic O–C passed all three internal tests, and it is a member of D9, the cleanest set this paper has.** Only two members of D9 have an external record long enough to test the secular reading, V527 Dra and V564 Dra (Sect. 5.5); one fails and the other agrees only within our window, and the other seven are untested by the literature. What does correct this false positive is internal: refitted with the complete TESS block of 40 sectors in place of four, the coefficient of V527 Dra falls from -0.0370 ± 0.0033 to between -0.0066 ± 0.0071 and $+0.0011 \pm 0.0264 \text{ s yr}^{-1}$ depending on the noise model — compatible with zero in all of them, which is what the published orbit implies (Sect. 5.5). The design saw a secular curvature where the complete block sees none; that is the mechanism of this section demonstrated end to end on the one system where the answer is known independently. Two earlier readings of this target — a “blind validation” of a fixed $\chi^2_{\text{red}} \geq 2$ flag, and a false-positive rate of “one in nine verified” — are withdrawn (Appendix D).

The mechanism is measurable. Six epochs over 15.7 yr sample a 2.73-yr cycle at essentially three phases (2008.4–2008.6, 2019.9–2020.1, 2024.1), and a three-parameter parabola passes through three groups, leaving only the intra-group structure as residual: this is why the goodness-of-fit test does not reject the parabola ($\chi^2_{\text{red}} 2.03$ against an expected median of 0.79 at 3 degrees of freedom — within the tail), whereas a 6.6-min signal against sub-minute bars would give χ^2_{red} of tens if the cycle were not absorbed. The published orbit (circular, $A = 6.6 \text{ min}$, $P_3 = 2.733 \text{ yr}$) was injected on an exactly linear ephemeris at our six epoch times with our bars, over 360 orbital phases (src/v527_ltte.py; expectation written before running; Fig. 5). The quadratic coefficient ranges from -0.058 to $+0.058 \text{ s yr}^{-1}$ ($|dP/dt| \geq 0.036$ at 57% of phases), $p_{\text{curv}} < 0.05$ at 93% of phases, $p_{\text{adv}} < 0.05$ at 84%, χ^2_{red} median 2.83 (below 3 at 52% of phases, maximum 5.6); 26 of the 360 phases return a coefficient within 0.005 s yr^{-1} of the observed one, with χ^2_{red} between 1.3 and 5.2 . The observed outcome is a typical product of a parabola over an undersampled third-body cycle. **The adversarial test shifts epochs, not phases, and does not protect against it: 84% of phases survive.** Any triple system in the sample with an outer period shorter than the baseline and longer than a sector passes

every internal test here and fails only an external one. The rest of this subsection asks how many such systems the sample is expected to contain.

The V527 Dra result turns a case into a rate question, with two ingredients that must not be confused: how often a close binary has a tertiary with an outer period inside the window where undersampling produces curvature, and how often such an orbit, sampled at a target's actual epochs with its actual bars, produces a detection by the tests of Sect. 3.5.

Detection efficiency per target. The 84% measured above is for one orbit and one sampling pattern and cannot be transferred. The efficiency was measured on each target (`src/epsilon_26.py`) by injecting a declared population of circular tertiary orbits — P_3 log-uniform in 0.1–20 yr, masses and inclination as in Appendix C, amplitude from Kepler's third law, which reproduces the published 6.6 min of V527 Dra — on an exactly linear ephemeris at the real epoch times with the real bars, 2 000 orbits per target, each judged by the curvature and adversarial tests. ε ranges from 0.00 to 0.53 (median 0.28, mean 0.25), is set by the size of the bars and not by the sampling pattern (Appendix C), and sums to $\Sigma\varepsilon = 6.4$ over the 26 with the bars of Sect. 3.3 (6.1 with the TESS bar of the previous version and 6.7 with the bars before the composition correction; 21.6 is the ceiling for orbits all as favourable as V527 Dra's). It treats orbit and secular term as exclusive; it covers circular orbits with $P_3 \leq 20$ yr only — orbits longer than the baseline, which are the ones most degenerate with a quadratic over 16–20 yr and have the larger amplitudes, are outside the injected population and outside the in-window fraction, so nothing below counts them; the fraction itself comes from solar-type spectroscopic binaries and is transferred to a sample dominated by contact and semi-detached systems; for the question of how many members of D9 — detections whose quadratic is also adequate — could be such orbits, the relevant efficiency is ε_3 , that of a detection passing the goodness-of-fit test as well, which sums to $\Sigma\varepsilon_3 = 4.2$ over the 26 rather than 6.4; and it is measured with the bars held fixed: an orbit that spreads a target's seasons also inflates the bars the pipeline would co-estimate from them (Sects. 5.2.4 and 5.2.5), so the pipeline's own efficiency is at most this one — the archival-lever part of ε shrinks under re-estimation, the TESS-internal part, whose bars come from half-sectors, does not — and $\Sigma\varepsilon$ and every number built on it below are ceilings. Two targets (TIC 224605072, 229476285) have $\varepsilon = 0$: their SuperWASP bars are too large for the curvature test to reject anything.

In-window tertiary fraction. From Table 5 of Tokovinin et al. (2006; VizieR J/A+A/450/681), of the 42 solar-type spectroscopic binaries with inner period below 3 d, **8 have the tertiary of the inner pair itself at an outer period of 0.1–20 yr**: $f = 0.19$, 95% Wilson interval 0.10–0.33. The count is the strict one — the companion has to be the tertiary of the eclipsing pair, not any component of the system; the looser count used up to the previous version, and what it added, are recorded in Appendix D. In the two longer bands the strict counts are 0 of 42 at 20–50 yr and 3 of 42 at 50–100 yr. The TESS residuals of this sample give an independent bound, with the estimator matched to the measurement (Sect. 5.2.4): the declared population is compatible with the strict fraction and with values well above it (probability 0.57 of a TESS factor as low as measured at $f = 0.24$), and only $f \approx 1$ is excluded at 95%. The internal test is one-sided and bounds f from above only, which is why the central value is taken from Tokovinin et al. and not from it. Survey-detected fractions (Borkovits et al. 2016; Hajdu et al. 2019; Marcadon & Prša 2024) are floors already convolved with each survey's efficiency and are not multiplied by ε (Appendix C). The transfer

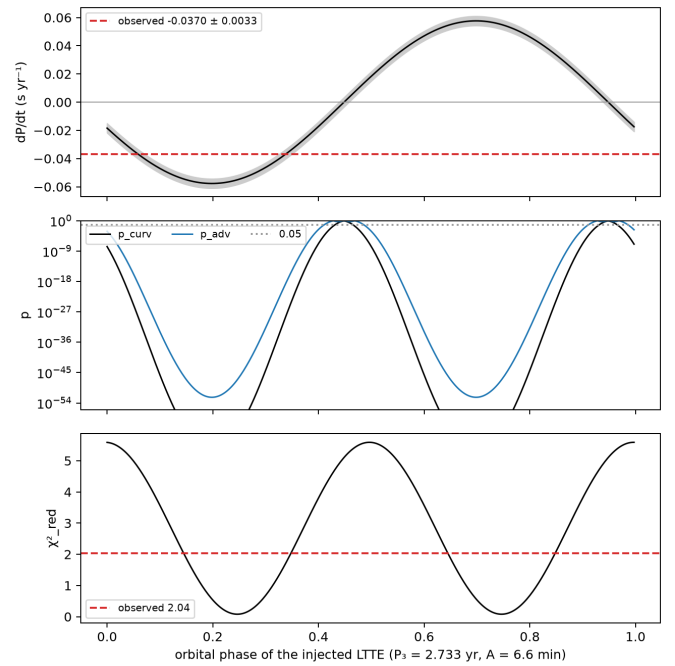


Fig. 5. The published LTTE orbit of V527 Dra injected on a linear ephemeris at our six epochs, over 360 orbital phases: the resulting quadratic coefficient (top), the curvature and adversarial p-values (middle) and the reduced χ^2 (bottom) against phase, with the values our chain returns for the target marked ($dP/dt = -0.0370 \pm 0.0033$ s yr $^{-1}$, $p_{\text{gof}} = 0.11$).

to this sample — six targets with $P > 2$ d, five detached — is asserted, not demonstrated.

The number. Two refinements of this count were made after the version that first reported it, and both are included here. The tertiary fraction is taken from the same Table 5 of Tokovinin et al. (2006) but counted strictly — the closest tertiary of the *inner* pair, not any pair of the system — which gives 8 of 42 in 0.1–20 yr rather than 10; and the injected population is extended to outer periods of 20–50 and 50–100 yr, the regime most degenerate with a quadratic over 16–20 yr, with the strict fractions 0 of 42 and 3 of 42. The efficiency is re-measured with the orbits injected on top of the *same* correlated noise as the between-sector model of Sect. 5.3.2 and judged by the same tests: $\Sigma\varepsilon_{\text{gp}}$ is 3.88, 13.67 and 7.19 in the three bands, 0.66–0.83 of the values with fixed bars. The false-alarm rate of the injection in that metric is zero — no realisation without an orbit passes the gate in any target; the summed rate over the 26 is 0.003 target, and the 95% upper bound implied by 0 of 500 realisations per target is 0.16 target in all. The expected number of detections produced by undersampled orbits is then $N = 1.25$, with a 95% Jeffreys interval of 0.80–2.46 over the three bands (with fixed bars instead of the correlated noise the same sum is 1.88, of which 1.11 comes from the 0.1–20 yr band; 0.91 for detections that would also pass the goodness-of-fit test), against the seven designs that survive the noise model of Sect. 5.3.2 — that is, about one in six of them. One of the eleven is a known such orbit (V527 Dra). For TIC 232634196 a third body is not excluded: $f \times \varepsilon$ is 0.02 under the declared population and 0.2 for a V527 Dra-like orbit.

5.5. The window test: the same coefficient from the complete TESS block, and the external records

The three mechanisms above are internal. This section confronts each design with two things it did not use: the complete TESS block of the same target, and the published record where one exists.

The test. For the nine members of D9 the coefficient was re-fitted to every primary epoch the cache holds — 38 to 43 sectors over 2019.6–2024.9 for eight of them, against the four the design uses — under each of the noise models of Sect. 5.3.2, and compared with the tabulated coefficient as $z = (dP/dt_{\text{TESS}} - dP/dt_{\text{design}})/\text{the quadrature sum of the two bars}$. One target, V Gru (TIC 144194304), has only four cached sectors and is not testable. Classes follow a rule fixed before the comparison was read: *contradicted* if $|z| > 3$ under every admitted model, *not contradicted* if $|z| < 2$ under every admitted model, *ambiguous* otherwise, where admitted has the meaning of Sect. 5.3.2. V527 Dra is held out of this classification, because its published light-travel-time orbit is precisely a signal the TESS block would have to fit, and it is reported instead by the complete fit (Sect. 5.4). The classification does not depend on the shared epochs: repeating it with the four design epochs removed from the block — leaving 35 to 39 epochs that share nothing with the tabulated fit — changes no class, moves no z by more than 0.09 and no bar by more than 1%.

The result is that the block contradicts four of the seven coefficients it classifies and agrees with power in one. Seven of the eight testable targets are classified (V527 Dra is held out), and four of the seven are contradicted:

- **TIC 392536812** — block $+0.021 \pm 0.007$ under its three admitted models against a tabulated $+0.053 \pm 0.003$: $z = -4.1$. (The pooled-GP variant listed in Table 5 has a bar three times larger and does not reach 2σ ; it is not an admitted model for this target.)
- **TIC 198408416** — $+0.038 \pm 0.005$ against $+0.010 \pm 0.001$: z from $+3.4$ to $+6.1$.
- **TIC 390021728** — $+0.029 \pm 0.009$ against -0.005 ± 0.001 : z from $+3.8$ to $+5.5$, and the block gives the *opposite sign*.
- **TIC 232634196** — -0.65 ± 0.13 against -0.217 ± 0.021 : z from -3.2 to -9.5 .

Two are not contradicted — TIC 230386284 ($+0.0091 \pm 0.0008$ against $+0.0081 \pm 0.0006$, $z = +1.0$) and TIC 377253090 (-0.005 ± 0.005 against -0.0063 ± 0.0010 , $z = +0.3$) — and one is ambiguous, V564 Dra (TIC 329246824), whose only admitted model gives $z = +2.9$. V527 Dra, held out, has a complete-block coefficient compatible with zero in every model (-0.0066 ± 0.0071 to $+0.0011 \pm 0.0264$) against a tabulated -0.0370 ± 0.0033 . Widening or narrowing the admission interval to 99% or 90% changes no class.

What “not contradicted” is worth. The minimum difference the block could have detected at 3σ under the admitted model is 0.0024 s yr^{-1} for TIC 230386284 and 0.0146 for TIC 377253090. The first is a third of the tabulated coefficient: the complete block agrees with the design there **at a power that would have shown a disagreement of a third of the value**, which is the strongest agreement in the sample. The second is more than twice the coefficient, so that target is untested rather than confirmed. For the ambiguous and the contradicted targets the same quantity runs from 0.020 to 0.40 s yr^{-1} . **No target is confirmed as a secular rate by this test:** agreement within the achievable power is the strongest statement any of them supports.

What the block shows, and what it does not separate.

Grouped by calendar year against the quadratic fitted to everything, three of the four contradicted targets show a wander of 2.4 to 8.5 min (TIC 198408416 2.4, 390021728 2.6, 232634196 8.5) against within-year dispersions of 0.2 to 2.2 min and bars of 0.1 to 0.8 min; the excursions peak in 2021–2022 and return, and in TIC 232634196 a cubic term is required ($\Delta\chi^2 = 39$ for one parameter, $p \approx 4 \times 10^{-10}$). **The fourth contradicted target is contradicted by the value of its coefficient and not by a wander:** TIC 392536812 moves by 0.82 min over the six years, against 0.11 and 0.66 min in the two uncontradicted targets and 1.49 min in the ambiguous one, so the size of the excursion does not separate the classes — a target can disagree with its own tabulated coefficient while its block is quiet. V527 Dra, held out, shows 8.2 min. Nor is the wander a higher-order polynomial: outside TIC 232634196 a cubic term adds $\Delta\chi^2 \leq 3.5$ for one parameter in every target, so what the block sees is structure on a scale of years that no polynomial of the degrees tested describes. This is the same statement as Sect. 5.3.2 seen in the time domain: what the design reads as curvature over 16–20 yr is, in the targets where the block can see it, a wander of minutes over years.

The external records. Published work on the 26 was reviewed by position (SIMBAD 5'': 26 of 26 matched; the VarAstro catalogue 1': 13 of 26 have an entry, 6 have ≥ 4 CCD primary minima spanning 7.7–34 yr). No target has a published dP/dt ; five have object-scale papers, of which one, V527 Dra, has a published light-travel-time orbit and no secular term (Sect. 5.4). Two targets have an external record long enough to test a secular reading, and they say the same thing as the block. **CV Dra** (TIC 229687624) has 83 CCD minima over 1988–2022: a quadratic over the whole record gives $+0.0088 \pm 0.0005 \text{ s yr}^{-1}$ against our -0.0014 ± 0.0021 ($z = -4.7$), while the same fit restricted to our own window gives -0.0003 ± 0.0017 ($z = -0.4$). Its O–C runs from +5 min in 1988 through –10 min in 2002 to +8 min in 2022, with quadratic residuals reaching 9.6 min against an intra-year scatter of 2.4 min (Fig. 2). **V564 Dra** (TIC 329246824) has 10 CCD minima over 1999–2025: $-0.0066 \pm 0.0011 \text{ s yr}^{-1}$ over the whole record against our -0.0207 ± 0.0026 ($z = -5.1$), and -0.0130 ± 0.0042 within our window ($z = -1.6$) — the same sign, a third of the size, and the disagreement is with the window and not with the measurement. It is also the target the complete TESS block calls ambiguous. At the level of the data rather than of the coefficients, our epochs placed on the VarAstro ephemeris next to CCD minima taken within a year agree in 2 of 2 comparisons in the SuperWASP era (differences –1.2 and +2.7 min, rms $z = 0.80$) and in 12 of 13 in the TESS era (rms $z = 1.17$); for V391 Dra (TIC 237116051) the cross-match is not usable, because VarAstro adopts half the catalogue period and 5 of its 7 “primary” minima are secondary minima under the convention of Table 3.

TIC 232634196, and the prediction that decides it. TYC 3906-474-1 (Tmag 9.9, $P = 1.2511 \text{ d}$) has the largest tabulated coefficient of the sample, $-0.217 \pm 0.021 \text{ s yr}^{-1}$, and it is the one detection carried by the archival lever: on the 20-yr linear ephemeris of the primaries the three SuperWASP seasons sit at –137, –100 and –110 min with 19.9-min bars, and the quadratic through them predicts the change seen inside the TESS block of the design. Its secondary moves in the same direction as its primary ($-2.4 \pm 0.9 \text{ min}$ against $-11.2 \pm 1.5 \text{ min}$ between 2019–2020 and sector 75), which excludes the anti-correlated signature of apsidal motion; the cycle count across the gap is unambiguous ($\sigma_p \times N = 0.002$); there is no VarAstro record and no literature on the star.

The complete block does not sustain the secular reading. Fitted to the 43 cached sectors alone, the coefficient is $-0.65 \pm 0.13 \text{ s yr}^{-1}$ under the admitted model (-0.70 ± 0.05 under the pooled one), three to nine sigma from the tabulated value; fitted to everything, -0.29 ± 0.02 with a reduced χ^2 of 4.1; and the year-by-year O–C of the block against that fit runs -2.1 , $+4.9$, $+2.9$ and -1.9 min in 2020, 2021.5, 2022–23 and 2024, an excursion of 8.5 min that a cubic term improves by $\Delta\chi^2 = 39$ for one parameter. The local period of 2024, from the eleven sectors 76–86, is $+0.46 \pm 0.44 \text{ s}$ away from the mean period of the ladder, where the quadratic fitted to everything requires $-0.75 \pm 0.06 \text{ s}$: a 2.7σ conflict, and the local ephemeris agrees with the linear one instead.

Instead of developing the object further the paper publishes the prediction that separates the ephemerides. The star (RA 17h58m, Dec $+56^\circ$) is near solar conjunction in January and observable from May to August, so the prediction is given for the observing season: for cycle E = 2206, the primary minimum nearest 2027.5, the quadratic fitted to everything gives BJD_TDB 2461588.806 $\pm$ 1.8 min, the linear one 2461588.821 (+20.6 min later) and the 2024 local ephemeris 2461588.822 (+21.7 min, $\pm$ 6.5). Scaled by the square root of the reduced χ^2 of each fit — necessary, since the fits do not describe the data — the windows become ± 3.7 , ± 3.9 and ± 9.5 min and the separation between the linear and the quadratic prediction is 3.9σ . A single minimum timed to a minute in mid-2027 therefore decides between them, and the 2024 local ephemeris already sits on the linear side. **This is a test between two ephemerides, not a confirmation of either.**

A prediction registered in advance. When TESS returns to the northern field, the sectors after s86 will give the same test out of sample. The prediction is registered here by class and precedes any sector later than s86, because no such sector exists: the MAST inventory of Sect. 2 returns nothing after s86 for the eight, TESS having observed the south in 2025–2026. The script that will run the test when those sectors arrive, `ffi_fora_amostra_d9`, carries the prediction in its docstring in the public code repository, and the deposit archived at Zenodo (Data availability) carries the date stamp. For the targets *not contradicted* the quadratic fitted to everything should win over the 2024 local linear ephemeris, with χ^2/n below 2; for those *contradicted* the 2024 local ephemeris should win, and the 7-epoch quadratic should lose by the largest margin; for the target classed as ambiguous no prediction is made. A not-contradicted target in which the local ephemeris wins, or a contradicted one in which the design quadratic wins, refutes the reading of this section.

6. What is and is not claimed

Claimed:

- the coefficients of Table 3 and their distribution in the partitions of Table 4 — sign-symmetric, with a median $|dP/dt|$ that depends on the partition — for the subpopulation defined by Tables 1–2, **as quadratic coefficients over 16–20 yr and not as rates**;
- that with formal timing errors the same chain would have reported 24 of 31 targets as significant period changes;
- that with bars co-estimated from two or three samples the residual test has a null of 0.77, not 1; that the apparent archival excess of 2.73 was a misplaced $\sqrt{n}$ measured against that wrong null, and the remainder, 1.31, is located in three targets whose quadratic the goodness-of-fit test rejects; and that over the 23 targets with an adequate quadratic

neither archive shows excess against the null with the same cut applied inside it (SuperWASP 0.72, TESS 0.68), so no bar correction is applied — with the limit that the test does not separate correct bars from bars inflated by a real secular term of the tabulated size;

- that the completeness of the chain for a secular term, with the same estimator, is 50% at 0.014 s yr^{-1} , and that the co-estimated bars suppress the significance of a real term by a factor that grows with it (median 2.9 at 0.2 s yr^{-1} , up to 24 in the targets with the smallest archival bars), leaving the coefficient unbiased and the goodness-of-fit test blind;
- that the between-sector timing noise is measured, not assumed: a white excess of 0.42 min (median, against a quadratic) required at 95% in 17 of 21 informative targets, correlated with $\tau_c \approx 68 \text{ d}$ and an amplitude of 0.8 min pooled, and not a random walk at 2σ ; that under a noise model admitted by its own goodness of fit on the extended block, **7 of the 11 detections survive** (TIC 144194304, 198408416, 230386284, 232634196, 329246824, 377253090 and 392536812), of which three are contradicted by the complete block of Sect. 5.5; and that the pooled amplitude, which is what makes the error bar of the window test, is not calibrated in any of the eight targets where the complete block can test it (χ^2/ν from 0.05 to 6.4);
- that the complete TESS block, 38 to 43 primary epochs per target and 35 to 39 of them independent of the design when the shared epochs are removed, **contradicts four of the seven coefficients it classifies at more than 3σ under every admitted noise model** (TIC 198408416, 232634196, 390021728 and 392536812 — one of them with the opposite sign) and leaves one ambiguous; that of the two it does not contradict, TIC 230386284 agrees at a power of 0.0024 s yr^{-1} , a third of its coefficient, and TIC 377253090 is untested at its own power of 0.0146 s yr^{-1} ; and that none is confirmed as a secular rate;
- that V527 Dra, whose published O–C is a light-travel-time orbit with no secular term, passes the significance, adversarial and goodness-of-fit tests on the design and is corrected by the complete block, whose coefficient is compatible with zero;
- that undersampled orbits of the declared population, injected on the same correlated noise and judged by the same tests, account for $N = 1.25$ detections (95% Jeffreys 0.80–2.46), with a false-alarm rate of zero in that metric;
- that the archival timing bias is $-1.26 \pm 0.75 \text{ min}$ when calibrated against ephemerides that include ground-based timings; that it is not reproduced by the estimator acting on the transit shape (-0.10 min), does not come from the time-system conversion (0.002 s), and rests on one of the four calibrators; and that its uncertainty is a first-order term in the tabulated coefficients (more than 1σ in 16 of 26 between 0 and -4.28 min) and a zeroth-order term in the verdicts;
- that where an independent 34-yr record exists (CV Dra), the same window reproduces the coefficient and the whole record does not.

Not claimed:

- **that any tabulated coefficient is a secular rate.** None is confirmed: of the seven the complete block classifies, four are contradicted and one is ambiguous, one agrees at a power of 0.0024 s yr^{-1} and one is untested at 0.0146 ; V527 Dra is a published orbit and V Gru is not testable with four cached sectors;

- the functional form of any O–C diagram; the presence of third bodies in any system beyond the published one; the physical mechanism behind any individual coefficient;
- that the adversarial test has a significance level: it is a robustness gate (Sect. 3.5);
- that the noise model of Sect. 5.3.2 is unique. It is the one admitted by its own goodness of fit on the extended block, chosen by a criterion written after the models were compared and using χ^2/ν alone; the pooled alternative gives the same count with a different membership, and the choice changes which targets survive, not how many;
- any bound on the in-window tertiary fraction from the residuals of this sample: the TESS side excludes nothing below the highest published fraction (Sect. 5.2.4);
- that the TESS bars inflated by 1.0 min yr^{-1} are calibrated (they are a ceiling);
- that the chain bias transfers from four WASP planets to deep, short eclipses — it is carried as a declared limitation, bounded only by two archival minima of CV Dra ($n = 2$);
- that the tabulated $\sigma(dP/dt)$ is the measurement error of a real secular term: for a term of $0.1\text{--}0.2 \text{ s yr}^{-1}$ on designs with $1.5\text{--}5 \text{ min}$ archival bars it is an overestimate by up to an order of magnitude;
- that the timing of TIC 377253090 is free of its unresolved $1.7''$ companion of equal brightness: a constant companion does not move a minimum, but its variability was not tested;
- that a target not contradicted by the complete block is confirmed by it;
- that the contradictions identify the *origin* of what the block sees — the wander of minutes over years is not attributed here to spots, to a third body or to period change — nor that every contradiction comes from such a wander: one of the four (TIC 392536812) is contradicted by the value of its coefficient while its own block is quiet (Sect. 5.5).

The flagged targets are not developed as case studies here. Earlier readings withdrawn in the course of the work are listed in Appendix D.

7. Data and code

Catalogues: Prša et al. (2022; VizieR J/ApJS/258/16), Kostov et al. (2025; J/ApJS/279/50). Photometry: TESS (Ricker et al. 2015) SPOC 2-min light curves (Jenkins et al. 2016) from MAST; SuperWASP public archive (Butters et al. 2010) as served by CERIT-SC (<https://wasp.cerit-sc.cz>). Published timings: VarAstro star catalogue and O–C data (var.astro.cz; Paschke & Brát 2006), queried 2026-09-13; bibliographies from SIMBAD (Wenger et al. 2000) and catalogues from VizieR (Ochsenbein et al. 2000), both queried 2026-09-13. Published ephemerides for the timing-bias calibration: Ivshina & Winn (2022; VizieR J/ApJS/259/62), Kokori et al. (2023; J/ApJS/265/4) and Bouma et al. (2020). Software: astropy (Astropy Collaboration 2013, 2018, 2022), numpy, scipy, pandas.

8. Data availability

The code that produces every number in this paper is developed in a public repository (<https://github.com/rondinellewalitte/dpdt-26-binaries>) and archived at Zenodo under the concept DOI <https://doi.org/10.5281/zenodo.22740628>, which always resolves to the latest version. The version DOI 10.5281/zenodo.22740629 archives the code as

it stood before the revisions recorded in Appendix D (items 17–19); the version that matches this manuscript is deposited under the same concept DOI at submission, and a further version DOI will be minted at acceptance. The archive holds, besides the chain itself (cadeia_oc, oc_lote, superwasp, epocas_142874476, ids) and the table, figure and manuscript generators (tabela_nota, figuras_nota, epocas_cds_26, md_para_tex, numeros_da_nota, confere_nota), one script per measurement reported here, each with its expectation written in the docstring and committed before the run:

- Sect. 2 and the sample: vies_cache_288 (sector census), arquivo_completo_26 (MAST manifest of the 799 light curves), ffi_fora_amostra_d9 (product inventory after s86 and the FFI-versus-2-min control).
- Sect. 3 and the bias: run_hj_calib, vies_hj_indep (calibration against published ground-based ephemerides), vies_forma_swasp (shape test), checa_tempo_swasp (time-system check), sensib_vies_26 (sensitivity of every coefficient and verdict to the bias), barra_tess_metades_26, gls_vies_26, blending_26.
- Sect. 5.1–5.2: qual_barra_26, nulo_barras_reestimadas_26, nulo_pisos_26, vies_alavanca_26, estrutura_swasp_26, completude_secular_26.
- Sect. 5.3: secundarios_26, jitter_232634196, jitter_primarios_26 (799 sectors, σ_j and variogram), deriva_gp_26 (pooled and per-target Gaussian process), reinflar_tess_26 (ceiling), ruido_admitido_26 (admitted noise model and the class criterion), reconcilia_hi_d9 (the four noise models side by side).
- Sect. 5.4: epsilon_26, epsilon_longo_26 (long-period bands, strict tertiary count, injection on correlated noise), v527_ltte.
- Sect. 5.5: caso_232634196, caso_232634196_completo, caso_completo_d9 (the window test, with and without the design epochs), previsao_232634196, literatura_26, comparar_brno_26.
- State comparisons: comparar_estados_BD and comparar_estados_DE, which reproduce what each change of the chain moved (Appendix D).
- Kept for the record of withdrawn corrections: reinflar_swasp_26, reinflar_conjunto_26.

The per-target records (data/orquestra/oc_lote/oc_<TIC>.json) carry every epoch, bar and fit; the derived tables used here are tabela_26, caso_completo_d9, caso_completo_d9_sem_desenho, ruido_admitido_d9, ruido_admitido_26, reconcilia_hi_d9, sensib_vies_26, jitter_primarios_26, deriva_gp_26_hiper, deriva_gp_26_desenho, epsilon_longo_26, epsilon_gp_26, qual_barra_26, nulo_barras_reestimadas_26, nulo_condicionado_26, nulo_pisos_26, completude_secular_26, secundarios_26, comparacao_brno_26, sobreposicao_brno_26, literatura_26, vies_forma_swasp, vies_hj_indep, ffi_fora_amostra_d9 and the state comparisons comparacao_estado_BD and comparacao_estado_DE. The eclipse timings of the 26 targets are also given as a machine-readable table (reports/epocas_26.csv, 145 epochs) for CDS. numeros_da_nota writes every number quoted in this paper with the file it comes from, and confere_nota compares the two and fails if any disagrees.

Appendix A: Outcome of the recovery pass, per target

Of the eight targets whose period ladder had not closed in the main batch, intermediate 2-min sectors were fetched from MAST (96 light curves; 21–23 sectors each for the four in the northern continuous-viewing zone, 1–3 for the others) and the batch was rerun. Of those eight, six still fail the ladder in the rerun, with median half-sector bars of 1.0–4.1 min (the tolerance is three times the median bar, 3–12 min here) that the cycle count cannot bridge without an outlier-sector rule that does not exist. One more stays in the row for a different reason: TIC 128790976 closed its ladder with the added sectors under the earlier TESS bar and then failed phase coverage, and with the bar of Sect. 3.3 its tolerance shrinks to 6.4 min, so the ladder no longer closes and the target remains in row 4, where the coverage failure would meet it next. One leaves: TIC 353871287 now stops earlier, at the 500-point floor of its SuperWASP curve (row 1). And one enters from outside the row: TIC 159638876, whose run had crashed in the main batch before the ladder — the supplementary manifest was read twice, the duplicated sectors had zero spread and an undefined weight, and the failure was listed then as a catalogue period error — now reaches the ladder and fails it with a median half-sector bar of 19 min. Six plus one, minus one, plus one: **row 4 holds eight targets in the rerun**, the same eight as in Table 2 (TIC 128790976, 159638876, 166606391, 206544316, 353939279, 356896550, 359675849 and 469979014).

Of the six without an applicable eclipse duration (row 5), one is a contact binary (morphology 0.99, catalogue width 36.6 h for $P = 4.8$ d) that the trapezoid does not describe and was not attempted. For the five others the eclipse width was measured on the folded TESS curve; for two of them the estimator returned exactly its 35%-of-period ceiling. That value is not a measurement of T_{14} — it is the estimator failing to locate a minimum when the curve is folded on the *catalogue* period, which points to the catalogue period rather than to the duration. These two are recorded as “minimum not found at the published period”, a different statement from “duration not published”, and one that would send a follow-up to the period, not to the eclipse. None of the six was recovered.

Appendix B: Control target and period-ladder audit

Before any new target was measured, the chain was run as a single function on TIC 142874476 and compared, component by component, with its previously recorded result (38 components: epochs, errors, residuals, χ^2 , Q , the SuperWASP source, cycle count, seasons, coverage status and the final residuals). Thirty reproduced to the quoted precision; the eight that did not were all downstream of one number — a quadratic extrapolation to 2007 of +38.9 min that existed only in a comment and that no fitting convention reproduces. The record was corrected to what the code produces. The adversarial test on the control gives $p = 0.28$ for the TESS-internal curvature (0.003 before the shift): with four epochs and one degree of freedom that detection is fragile, which is why the robust statement about the control is its 10σ departure from a constant period, not the curvature. The control was re-run before and after every change reported in this version and reproduced each time; under the chain bias of this version its archival epoch sits 21.8 min from the linear ephemeris (9.8σ) and 6.4 min from the quadratic (2.9σ).

Period ladder. The ladder returns epochs and residuals in the caller’s order. This was a latent defect: an earlier version returned them in internal time order, and unsorted input produced cycle

numbers assigned to the wrong sectors — caught only because the resulting dP/dt of $-24\,122\text{ s yr}^{-1}$ is physically impossible; a milder scramble would have looked like mass transfer. The $|dP/dt| > 10\text{ s yr}^{-1}$ guard of Sect. 3.2 exists because of it. The tolerance rule of Sect. 3.2 (the larger of 3 min and $3\times$ the median epoch uncertainty) was introduced after the main batch, which ran with a fixed 3 min; re-running all 26 targets under the rule reproduces every tabulated value (1 024 fields compared across the 26 per-target records; only the wall-clock time differs).

Appendix C: The orbit population, the efficiency, and the occurrence numbers

Population. Circular tertiary orbits with P_3 log-uniform in 0.1–20 yr (between a sector and the baseline), binary mass uniform in $1.5\text{--}2.5\,M_\odot$, tertiary mass uniform in $0.1\text{--}1.0\,M_\odot$, isotropic inclination ($\cos i$ uniform), phase uniform; the light-travel-time semi-amplitude follows from Kepler’s third law, $A = a_3 M_3/(M_{\text{bin}} + M_3) \sin i / c$, and reproduces the published 6.6 min of V527 Dra for a $1\,M_\odot$ tertiary at 70° and $P_3 = 2.733\text{ yr}$ with $M_{\text{bin}} = 2.5\,M_\odot$, the upper edge of the prior ($2.0\,M_\odot$ would give 7.3 min); the binary mass used in that check is therefore the prior’s edge, not a measured value. The median amplitude of the population is 2.0 min, comparable to the epoch bars. ε_3 (detection and $p_{\text{gof}} \geq 0.05$) sums to 4.2.

What sets ε . The expectation that ε would track the sampling pattern was not met: the Spearman correlation of ε with the number of epoch groups is -0.07 ($p = 0.72$) and with N is $+0.19$ ($p = 0.36$), while with the median bar it is -0.66 — the bars set the efficiency, not the pattern. Two targets have $\varepsilon = 0$ (their archival bars make the curvature test powerless at these amplitudes). For orbits as favourable as V527 Dra’s, ε approaches 0.83; that one empirical anchor is biased, since the orbit was published because it is large enough to solve from 1 832 timings.

Occurrence numbers and what each measures. Occurrence over all outer periods: 96% of solar-type spectroscopic binaries with $P < 3$ d after correction for incompleteness, 79% raw (Tokovinin et al. 2006); a lower limit of $42 \pm 5\%$ for contact binaries (Pribulla & Rucinski 2006). Detected fractions in eclipse-timing surveys: 8.5% of $\sim 2\,600$ Kepler binaries with outer periods below $\sim 2\,000$ d (Borkovits et al. 2016); 1.2% of $\sim 80\,000$ OGLE-IV binaries (Hajdu et al. 2019); 10 in ~ 100 TESS full-frame-image targets (Marcadon & Prša 2024). The detected fractions are convolved with each survey’s baseline, cadence and precision; multiplying them by ε applies an efficiency twice. The in-window fraction of Sect. 5.4 comes from the 33 tertiaries listed for the 42 $P < 3$ d systems of Tokovinin et al.’s Table 5, counted strictly — the tertiary of the inner pair itself, which gives 8 of 42 in 0.1–20 yr; the two systems that the loose count adds (HIP 16846 and 61910) have their in-window companion in another subsystem. All eight are spectroscopic or visual-orbit detections rather than resolved wide pairs.

Appendix D: Record of readings withdrawn between versions, and of the changes of state

Changes of state of the chain, in order. (A) The first batch used the standard error of the season mean as the bar of each SuperWASP season, $\sqrt{2} - \sqrt{3}$ too small in 49 of 61 epochs; the residual factor of the archive was 2.73 and the median reduced χ^2 1.41. (B) Corrected to the season dispersion: the factor fell to 1.35, the median reduced χ^2 to 0.67, one detection was lost (TIC 359552377), and no coefficient moved by more than 0.04σ . (D)

The TESS bar from the halves of a sector was $|t_a - t_b|/\sqrt{2}$ — the dispersion of one half given to the whole sector — and became $|t_a - t_b|/2$: 36 of 84 bars shrank by a median factor of 0.71, one detection was added (TIC 390021728, p_{adv} 0.10 $\rightarrow$ 0.013), the TESS residual factor rose from 1.02 to 1.59, and no coefficient moved by more than 0.20σ . (E) The archival bias was recalibrated against ephemerides that include ground-based timings and went from -2.14 ± 0.78 to -1.26 ± 0.75 min (Sect. 3.4); every archival epoch moved by $+0.88$ min, the 26 coefficients all moved in the same direction by a median of 0.29σ (maximum 0.95σ), the median $|dP/dt|$ went from 0.016 to 0.014 s yr $^{-1}$, and no verdict, class or flag changed. Each state was run against an expectation committed before the run, and the control target reproduced in all of them.

Readings withdrawn.

Each entry names the reading, the version that carried it, and what replaced it. They are kept here rather than in the body because the body is about the data; they are kept at all because each was, at the time, the honest reading of a number that turned out to measure something else. They fall into three groups.

Errors in the estimator or in what it was compared with.

1. *A median reduced χ^2 of 1.41 read as bars 1.39 times too small*, and the archival variance factor of 2.73 read as *SuperWASP bars 1.65 times too small*, with a “corrected” set of ten detections and a “conservative joint correction” (SuperWASP $\times 1.65$, TESS $\times 1.34$) leaving eight. The composition error of Sect. 5.2.2 — the standard error of the season mean assigned to each season as its bar — accounted for a factor of 1.74; corrected, the factor is 1.35 and the median 0.67, and the corrections are withdrawn.
2. *The remaining excess as small-sample bias*. The expectation written for the null test of Sect. 5.2.3 was a SuperWASP null of 1.3–1.8 from the $\nu/(\nu - 2)$ argument; the null is 0.77, because r and s are the same numbers, and the measured 1.35 is an excess located in the three targets with a rejected quadratic.
3. *The 23 adequate targets against an unconditioned null*. A version compared the residuals of the 23 targets that passed the goodness-of-fit cut with a null simulated on the same 23 designs without the cut (0.78; TESS 0.96, the measured 0.55 at its 13th percentile). With the cut applied inside the null the TESS null is 0.70 and the measured value is at the 23rd percentile (Sect. 5.2.3); the same conditioning was propagated to the injection of Sect. 5.2.4, where the 23 had likewise been compared without the cut.
4. *“ $z^2 = 0.5$ by construction” as the reason for the null of 0.77*. A version explained the null of Sect. 5.2.3 by the tied residual and bar of a two-season pair; the leverage-corrected ratio of such a pair is exactly 1, and the null below 1 comes from the floors on the bar (an external check; Sect. 5.2.3).
5. *A TESS-side bound of $f \lesssim 0.5$ on the in-window tertiary fraction*, and the reading that orbits at $f = 0.24$ explain 42% of the archival excess, both obtained by injecting with the bars held fixed and comparing with a measurement whose estimator re-estimates them. Replaced by the injection with the matched estimator, whose TESS-side bound ($f \gtrsim 0.8$ excluded) is uninformative, and an archival side nearly blind to orbits (Sect. 5.2.4).

Transfers from one object or one population to another.

6. *A borrowed efficiency*. A version multiplied survey-detected tertiary fractions by an efficiency of 0.83 taken from the V527 Dra injection; the fractions are already convolved with

each survey’s efficiency, and the efficiency of one orbit at one sampling pattern does not transfer. Replaced by the per-target ε on a declared population and the in-window fraction of Tokovinin et al. (Sect. 5.4).

7. *“One in nine verified”*. A version quoted a false-positive rate of one in nine against the literature; seven of the nine had no record long enough to test. Replaced by an existence proof and an injection rate (Sect. 5.4).
8. *The residuals as a bound on the transferred chain bias*. A version stated that the SuperWASP residuals of the 23 adequate targets, showing no excess, left no room for a shape-dependent bias larger than the bars. A bias common to a target’s seasons is a block offset that the quadratic absorbs; the residual test cannot see it, and the only bound is the two CV Dra minima of Sect. 5.5 (Sect. 3.4).

Criteria that measured something other than what they were named for.

9. *A fixed flag $\chi_{\text{red}}^2 \geq 2$ as a validation*. An early version flagged 7 of 26 targets by a fixed reduced χ^2 and read the flag on V527 Dra as a blind validation; the flag’s false-alarm probability depends on the degrees of freedom (0.157 at 1, 0.092 at 4), a coincidence with 7 of 26 flagged had a 27% prior probability, and the criterion was replaced by $p_{\text{gof}} < 0.05$ (Sect. 3.5).
10. *TESS bars “calibrated” at 1.0 min yr $^{-1}$* . A version called the TESS bars inflated by the secondary drift calibrated because they brought the median reduced χ^2 to 0.73; the coincidence came from re-weighting, and the inflation is a ceiling (Sect. 5.3.2).
11. *Eleven detections and nine adequate ones*. TIC 359552377 left the set when the composition error was corrected (Sect. 4; p_{adv} 0.05 $\rightarrow$ 0.22). Its departure has two readings that the data do not separate: a spurious detection removed by the corrected bars, or a real term of the size it carries (-0.25 ± 0.08 s yr $^{-1}$) that the gate on this design — five epochs, archival bars of 16 min — passes in 26% of injections (Sect. 5.2.5). The count is the same under both; what it means is not.
12. *Numbers carried over from the version before the composition correction*. An external check found, after that correction, the coefficients of Sect. 5.5 (HD 202042, CV Dra, V564 Dra) and the z of the VarAstro comparison still at their earlier values, a Table 5 sentence with a target count and a p_{curv} path that its own table did not give, “1.0 and 2.6 min yr $^{-1}$ ” called the extremes of a drift whose extremes are 0.67 and 2.59, a Table 2 one batch behind the records, and a claim that the calibrators’ dispersion guard was applied to the targets; each is corrected in place (Sections 2, 3.4, 5.3.2, 5.5, Appendix A).
13. *A TESS bar too large by $\sqrt{2}$, and the sets of ten and eight built on it*. Up to the previous version the halves term of the TESS bar was $|t_a - t_b|/\sqrt{2}$, the dispersion of one half given to the whole sector (Sect. 3.3); an external check found it, a simulation with an independent truth confirmed it, and the chain and every test of Sect. 5 were rerun with $|t_a - t_b|/2$ (expectation written and committed before rerunning: the control unchanged, the same 26, one detection added, the TESS residual factor 1.02 $\rightarrow$ 1.59, the TESS null of the 26 rising to 1.1–1.6). The control reproduced; the same 26 were measured; TIC 390021728 entered the detected set (p_{adv} 0.10 $\rightarrow$ 0.013) and none left, so D10 and D8 became D11 and D9;

the median reduced χ^2 moved from 0.67 to 0.70 and the aggregate from 1.20 to 1.47, which is above the co-estimated null where 1.20 was at its 93rd percentile — the two flagged targets that hold the TESS excess (Sect. 5.2.3) gained χ^2 as their TESS bars shrank; the TESS residual factor rose from 1.02 to 1.59 and its null from 0.96 to 0.98 (below the expected 1.1–1.6: the TESS null is set by the formal-error floor, which the smaller halves term reaches more often — `src/nulo_pisos_26.py` gives 1.01 with that floor alone under either divisor), so the TESS side of the 26 went from no excess to a mild one at the 94th percentile, and over the 23 adequate targets from the 23rd to the 24th percentile of its conditioned null; a sixth secondary drift became significant (TIC 115244268); $\Sigma\epsilon$ rose from 6.1 to 6.4; the 50% completeness moved from 0.015 to 0.014 s yr^{-1} ; and one target of Table 2 moved from the coverage row to the ladder row (Appendix A). Every number of the earlier bar that the text still quotes is labelled as such. The null simulations could not have found this: they generated the halves and re-estimated the bar under the same rule, and returned 0.96 for a bar that was too large by $\sqrt{2}$.

14. “The eleven detections are carried by the TESS timings, all but one vanishing under bars inflated by the secondaries’ drift.” Superseded rather than withdrawn: that inflation is a ceiling, and the noise model now applied to the design is the one admitted by its own goodness of fit on the extended block, under which 7 of the 11 survive (Sect. 5.3.2).
15. “TIC 232634196 has a curvature of -0.215 s yr^{-1} over 16–20 yr.” The coefficient stands as a coefficient of that window; the secular reading does not: its own complete TESS block gives $-0.65 \pm 0.13 \text{ s yr}^{-1}$ over 2019–2025 and an 8.5-min excursion that no parabola describes (Sect. 5.5).
16. “At most one or two of the eleven detections are undersampled orbits ($f \times \Sigma\epsilon = 1.5$)”, with the in-window tertiary fraction counted loosely (10 of 42). Replaced by the strict count of the tertiary of the inner pair (8 of 42), the long-period bands and the injection on correlated noise: $N = 1.25$ (95% Jeffreys 0.80–2.46; Sects. 5.4 and 5.2.4).
17. The chain-bias equation printed with the value of the previous state. Up to the previous version the display equation of Sect. 3.4 was written into the converter rather than derived from the manuscript source, so the text read $-1.26 \pm 0.75 \text{ min}$ and the equation -2.14 ± 0.78 ; the converter now takes the equation from the same line as the text (an external check).
18. The figures of the PDF taken from a manual copy. The figure files the manuscript includes live in `paper/figures/`, and until this version they were a copy made by hand in an earlier state of the chain, while `src/figuras_notas.py` wrote the current ones to `reports/figures/`. The PDF therefore carried figures of the state before the bar corrections together with text of the current state: the panels showed “D10”, “1 of 10” and “observed -0.0359 ± 0.0040 ” — labels that the script computes from the data, so it was the file that was old, not the code. It survived four versions because the text inside a figure is rasterised and no check of the manuscript or of the PDF reads it. `figuras_notas.py` now mirrors the two directories at every run and asserts they are byte-identical, `src/md_para_tex.py` refuses to convert if they differ, and the text drawn inside each figure is written to `reports/figuras_texto.json` and swept by the same checks as the manuscript (`src/confere_notas.py`). Same family as the previous entry: a number that the chain produces correctly, published from a copy that the chain does not write.

19. The letter of an appendix taken from its position instead of from its name. This appendix and the previous one were written in the manuscript source in the order E, D, and LaTeX numbers appendices by order of appearance: the first PDF of this version printed the residual-test appendix as D and this record as E. Every cross-reference still resolved to the right content, the compiler reported no undefined reference, and the thirty-seven checks of `src/confere_notas.py` passed — the printed letter was the only thing wrong, and nothing was looking at it. The converter now requires the n-th appendix heading to carry the n-th letter and stops before writing any TeX. Three defects of this version share one shape (items 17, 18 and this one): the display equation written into the converter, the figures taken from a copy made by hand, and the appendix letter assigned by position. In all three the chain computed the right number and the manuscript published something else, through an artefact the chain does not write. Each now has its own check, and since this version every check runs *before* the TeX file exists, so a manuscript that fails one is not produced at all.

Appendix E: The residual test: simulations and their expectations

This appendix holds the numbers behind Sect. 5.2: the grids of the simulations, the conditioning of the null, the mechanism that couples a bar to its own residual, and the expectation written and committed before each run. Nothing here is new — it is the working of the results quoted in the body.

Why the null had to be simulated. For s independent of r , $z^2 = r^2/s^2$ follows $F(1, \nu)$ with mean $\nu/(\nu - 2)$, undefined for $\nu \leq 2$; an external reader pointed out that with $\nu = 1-2$ the bootstrap width of the factors was the signature of that, and the expectation written and committed before the simulation was a SuperWASP null of 1.3–1.8, with the then-measured 1.35 inside its central interval. **The expectation was wrong in direction:** with the bars re-estimated as the pipeline re-estimates them the null is 0.77 (2.5–97.5%: 0.63–0.94; quartiles 0.72–0.81) and the TESS null 0.98 (median; mean 1.03; 0.54–1.79), both below 1. The reason is not the tied residual of a two-season pair — the leverage-corrected ratio of such a pair is exactly 1 — but the floors: removing them in the simulation (`src/nulo_pisos_26.py`) returns the SuperWASP null to 1.15 and makes the TESS null diverge, because a half-sector dispersion near zero makes z unbounded; the three floors together give 0.78 and 0.95. Treating the 0.75 min bias term as a common offset of the archival block rather than as per-season noise changes nothing (0.76). The values are properties of this design — 26 targets, 4–7 epochs each, bars co-estimated with $\nu = 1-2$ and floored by the formal error and the 0.75 min — and what transfers to another sample is the simulation, not the number.

The conditioned null. The 23 targets with an adequate quadratic are those that passed the goodness-of-fit cut, so the null they are compared with has to apply the same cut inside every realisation: p_{gof} of each target’s quadratic computed with the re-estimated bars, the targets below 0.05 removed, the factors formed over the survivors. Under the null the cut removes 0.76 targets per realisation — 2.9% per target rather than 5%, because the re-estimated bars deflate χ^2 — and three or more in 4.3% of realisations, so the number of flagged targets in the data is itself a mild excess. The conditioned nulls are 0.77 (0.64–0.92) for SuperWASP, unchanged, and 0.81 (0.46–1.22) for TESS, below the unconditioned 0.98: the realisations that fail the cut are those in which a TESS bar co-estimated from two half-sectors came out

small, and they hold 23% of the TESS Σz^2 of the null in 3% of the target-realizations. Under the TESS bar of the previous version the drop was larger (0.96 to 0.70), against an expectation of 0.85–0.93 written before that run. An earlier version compared the 23 with the null of the same designs *without* the cut (0.78), which is not the same experiment; the external reader pointed this out, and the same conditioning was propagated to the orbit injection below.

A secular term through the residual test. With a term injected in every target and the goodness-of-fit cut applied inside each realisation (`src/completeness_secular_26.py -fator`; expectation written and committed before running: a monotonic fall, about 0.70 at 0.02 s yr⁻¹), the SuperWASP factor is 0.76, 0.71, 0.59, 0.43 and 0.29 at 0.005, 0.01, 0.02, 0.05 and 0.2 s yr⁻¹, while the TESS factor stays at 0.81–0.85 — half-sector bars do not see the term and the parabola absorbs it exactly. With the bars held fixed the factor is 0.99–1.01 at every amplitude: the term lies in the model space and the whole effect is the re-estimation. The measured 0.72 of the 23 sits at the 34th percentile of the null in this run (24th in the run of the conditioned null; Monte Carlo scatter) and at the 58th of the distribution with 0.01 s yr⁻¹ in every target; $P(\leq 0.72)$ is 0.88 at 0.02 s yr⁻¹ and 1.00 at 0.05. The probability of a factor ≥ 1.31 is 0.001 with no term and 0.04–0.08 with 0.01–0.05 s yr⁻¹ in every target: a population of real terms of the tabulated size is not excluded as the source of the excess of the 26, although the three flagged targets remain the simpler account of where it sits.

The coupling between bar and residual. The reading first written for the fall of the factor — that the inflated archival bars hand the fit to the TESS block — was tested and is wrong: the mean leverage of the archival epochs stays at 0.41–0.42 across the grid. What the term breaks is the coupling between a bar and its own residual. With two seasons the bar is $|\Delta|/\sqrt{2}$ and the residual $\pm\Delta/2$ of the same between-season difference Δ ; with a term, the bar is built from the signal's between-season difference plus the noise's, and the residual — the parabola having absorbed the term — from the noise's alone, so that when the two nearly cancel the bar drops to its floor while the residual does not.

Orbits through the same estimator. The injection of the declared population of Sect. 5.4 into the 26 designs, with the bars re-estimated inside the simulation (`src/estrutura_swasp_26.py`; expectation written and committed before running: the null reproduced at $f = 0$, a SuperWASP factor of 0.95–1.10 and a TESS factor near 1.5 at $f = 0.24$), reproduces the null at $f = 0$ (0.77) and gives, at the occurrence $f = 0.24$ at which it was run — the loose count of the earlier version, so an upper bound — a SuperWASP factor of 0.85 (below the expectation) and a TESS factor of 1.68. The probability that orbits alone produce the measured SuperWASP factor of 1.31 is 0.002 at $f = 0$, 0.022 at $f = 0.24$ and 0.16 at $f = 1$; the TESS factor of 1.59 is as compatible with the declared population at the Tokovinin et al. fraction as with none, which is why the TESS side bounds nothing useful. The first run of this injection held the bars fixed, whose null is 1.00 against the measurement's 0.77; the numbers it gave are withdrawn as cross-estimator comparisons (Appendix D).

Completeness and self-suppression, in detail. Over the 23 targets that reach 50% completeness, the amplitude at which they reach it correlates with the median archival bar (Spearman +0.46, $p = 0.03$) and, more weakly and in the opposite sense, with the number of epochs (–0.26, $p = 0.23$); with 23 targets, and bar size and epoch count themselves correlated, the two are not separated here, and the three targets that never reach 0.5 are the three with the largest archival bars. Re-estimating the

bars costs 12 points of completeness at 0.02 s yr⁻¹ (median 0.86 against 0.98) and nothing from 0.05 upward. The suppression factor $\sigma(dP/dt)$ with re-estimated bars over σ with the true bars has a median of 1.03 at 0.02 s yr⁻¹, 1.19 at 0.05, 1.6 at 0.1 and 2.9 at 0.2, reaching 5–24 at 0.2 in the nine targets whose archival seasons have bars of 1.5–5 min at the longest lever; at the coefficients of Table 3 the factor for the eleven members of D11 is 0.90–1.70 (median 1.26), and for the TESS side alone 0.90–1.59. The goodness-of-fit cut does not see the term ($p_{\text{gof}} < 0.05$ in 2–4% of realisations at the median, at most 21% in one four-epoch design). Two expectations were committed before this run, the external reader's and the author's, and they differed on the size of the suppression; the measured factor exceeded both at the largest amplitudes.

What would break the coupling. The alternative to the present bar is not only the formal errors, which Sect. 5.1 shows reporting 24 significant targets of 31: a bar estimated from the nights or halves of each season rather than from the seasons themselves, a dispersion model pooled over the 61 seasons as a function of magnitude, period and depth, or a hierarchical shrinkage of the per-target dispersions would each break the coupling between the bar and the term. None is implemented in this version, and each is a natural next step.

Acknowledgements. An external reader of an earlier version asked whether the archival variance factor could be an artefact of its denominator; the literal check of that question found the composition error of Sect. 5.2.2 and led to the null test of Sect. 5.2.3, and the paper owes its central section to it. This research has made use of the SIMBAD database and the VizieR catalogue access tool, operated at CDS, Strasbourg, France; of the VarAstro database of the Variable Star and Exoplanet Section of the Czech Astronomical Society; of data from the TESS mission obtained from the Mikulski Archive for Space Telescopes (MAST), funded by the NASA Explorer Program; and of the SuperWASP public archive.

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